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Notes on Classical Electron Dynamics: Lorentz–Dirac Equation

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ABSTRACT

These notes reorganize, twenty years later, the classical-electron part of an undergraduate thesis written in 2006, and set it against the literature it was drawing on. The material is arranged as one line of argument. The runaway appears with Lorentz, survives Dirac's much better derivation unchanged, and is then attacked along the two branches Eliezer identified in 1947: change the coefficient of the radiation-reaction term, or raise the order of the equation. Each branch is followed on its own terms and scored against criteria fixed before either is on the table. Two things are recorded as they arise — where a result the 2006 text reached numerically had already been obtained analytically, and where a step it relies on is in tension with what is now known. No new results are claimed, and the twenty-years-on assessment is left for a later section.

1. THE LORENTZ EQUATION AND THE RUNAWAY PROBLEM

Conventions, fixed here and used throughout. Electrodynamic quantities are in Gaussian units. The factor c is kept explicit in formulae rather than set to one, and where it is convenient to drop it — in the tube integral of Sec. 2.1.2 and in the non-relativistic reductions of Sec. 5 — that is said locally and c is restored at the end. The metric signature is $(-,+,+,+)$, so that $u^\alpha u_\alpha = -1$. The world line is $z^\alpha(\tau)$ with τ the proper time, and $u^\alpha = dz^\alpha/d\tau$, $a^\alpha = du^\alpha/d\tau$; an overdot means $d/d\tau$ except in explicitly non-relativistic passages, where it means d/dt. Differentiating $u \cdot u = -1$ gives $u \cdot a = 0$ and then $u \cdot \dot{a} = -a \cdot a$, both of which are used constantly and neither of which is flagged again.

One number is worth fixing at the outset, because every section below uses it. The characteristic time of the electron is $\tau_0 = 2e^2/3mc^3$, which in Gaussian units is $6.27 \times 10^{-24}\,\mathrm{s}$. D. Zhang (2006) declares Gaussian units but evaluates this quantity without the factor $1/4\pi\varepsilon_0$, obtaining $10^{-33}\,\mathrm{s}$ instead. The corrected value is used here from the start, and nothing else is made of the discrepancy.

1.1. *Electrodynamics does not supply an equation of motion for the charge*

Classical electrodynamics rests on three pillars: the conservation laws, chief among them the conservation of charge, Maxwell's equations, and the Lorentz force law (Z. Zhang 1979). Maxwell's equations fix the field produced by a given distribution of charge and current. The Lorentz force law fixes the force exerted on a charge by a given field. Between them the two appear to account for everything an electrodynamical problem could ask for, and they do not: neither supplies the equation of motion of the charge itself.

The gap is structural rather than an oversight. To follow the motion of a charge one needs the force acting on it, hence the total field at its location, and the total field includes the field that the charge itself produces. Maxwell's equations say how that self-field is generated but not how it acts back. The Lorentz force law says how a field acts on a charge but is silent on whether the charge's own field belongs among the fields that act. Electrodynamics is complete as an account of how charges make fields and of how fields push charges, and it is incomplete exactly where the two meet on the same particle. Closing that gap is the subject of everything below.

1.2. *Three historical routes*

Any attempt to close the gap must begin by deciding what a charge is, and two pictures divide the subject between them. In the first the charge sits at a point. Its own field then diverges as the point is approached, and with the field the Lorentz force, so the force expression loses its meaning exactly where it is needed. One may object that the Lorentz force was only ever intended for external fields and that the self-field is simply not subject to it. But the objection buys nothing, because it leaves the self-force undetermined and the equation of motion no closer than before. That is the standing difficulty of the point picture. In the second the charge is spread over a finite region and treated as a continuous medium. Every quantity is then finite and the self-force is a well-defined integral over the distribution. The price is that like charges repel, so a body carrying charge of one sign must fly apart. Bound charges nevertheless exist, and electrodynamics does not say what holds them together.

Three attempts, spread over thirty years, took these pictures in turn. H. Poincaré (1906) worked in the extended picture and supplied the missing agent by hand, adding to the electromagnetic stress–energy tensor a binding tensor called the Poincaré stresses, chosen so that the total is in equilibrium and the charged body holds. The construction does what is asked of it, but it postulates a force for the sole purpose of removing a single contradiction, with no other consequence and no independent evidence for its existence. In the century since, few have followed it.

H. A. Lorentz (1909) took a position between the two pictures: the electron is a sphere of finite radius carrying a uniform charge. This is concrete enough to yield a definite equation of motion, which is the subject of Sec. 1.3.

P. A. M. Dirac (1938) returned to the point picture, but gave up computing the self-force as the mutual action of one part of the charge on another. He imposed momentum–energy conservation on a thin tube surrounding the world line and read the equation of motion off the balance, so that the divergent self-field enters only through quantities that cancel. This is the route the rest of these notes follow. It is set out in Sec. 2.

These notes adopt the point picture, for the reasons that recommended it to Dirac. It is technically uniform. The model carries no internal structure that has to be specified in advance. The equation of motion comes out in a fixed form. The construction also generalizes, both to curved backgrounds and to the corresponding gravitational problem. The cost is that infinite quantities appear at intermediate stages, so the force has to be extracted indirectly. This can be done either by balancing finite quantities, as in the conservation method, or by separating off a singular part and discarding it. In what sense those operations are legitimate is a question that runs through everything below.

1.3. *The Lorentz model and its three defects*

Lorentz's electron is a sphere of finite radius carrying a uniform charge. It is far enough into the extended picture that nothing diverges, and structureless enough to remain tractable. What holds the charge together is left unanswered, which is precisely the question the Poincaré stresses were invented to settle. In motion the sphere is contracted into an ellipsoid.

Two forces then act on it. One is external. The other is a self-force, produced by the action of the charge elements on one another. At rest this self-force vanishes by symmetry. In motion it does not, and the reason is retardation: the field with which one element acts on a second is the field it carried at an earlier time, and the reaction of the second on the first is delayed in the same way, so the two no longer cancel. The charge elements taken by themselves therefore violate Newton's third law. Nothing is genuinely lost. The missing momentum resides in the field. But the failure is a fair measure of how far the problem has already moved away from mechanics.

1.3.1. *Deriving the self-force directly*

The calculation itself is not deep. The work is all in the expansion, and it is standard (see, e.g., J. D. Jackson 1999; F. Rohrlich 2007). Let the charge be distributed with density ρ over a rigid body of size R, moving slowly enough that magnetic contributions may be dropped to the order kept. The self-force is the double integral of the retarded interaction over all pairs of charge elements,

$$\mathbf{F}_{\text{self}} = \iint d^3r\, d^3r'\, \rho(\mathbf{r})\rho(\mathbf{r}')\, \mathbf{K}(\mathbf{r}, \mathbf{r}'; t - s)\,, \qquad s = \frac{|\mathbf{r} - \mathbf{r}'|}{c}, \tag{1}$$

in which the kernel $\mathbf{K}$ is evaluated not at the present time but one light-crossing time s earlier. Since $s \sim R/c$ is small, expand every retarded quantity about the present instant,

$$f(t - s) = f(t) - s\,\dot{f}(t) + \frac{s^2}{2}\ddot{f}(t) - \frac{s^3}{6}\dddot{f}(t) + \cdots\,, \tag{2}$$

and collect. Each additional power of s costs one factor of R/c and buys one more time derivative of $\mathbf{v}$. The series therefore organizes itself by derivative order. The term containing $d^n\mathbf{v}/dt^n$ carries R^{n-2}. The zeroth term is the static self-repulsion, which vanishes by symmetry for a rigid symmetric body, and what survives is

$$\mathbf{F}_{\text{self}} = -\frac{4U_0}{3c^2}\,\dot{\mathbf{v}} \;+\; \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}} \;+\; \sum_{n\geq 3} \alpha_n\, \frac{e^2 R^{n-2}}{c^{n+1}} \frac{d^n\mathbf{v}}{dt^n}, \tag{3}$$

where U_0 is the electrostatic self-energy and the α_n are pure numbers fixed by the shape of the distribution.

The three groups of terms meet three quite different fates, set out in Table 1, and the whole of the subject descends from that split.

Term	Coefficient scales as	What becomes of it
$n = 1$, $\propto \dot{\mathbf{v}}$	R^{-1}, divergent	absorbed into the mass
$n = 2$, $\propto \ddot{\mathbf{v}}$	R^0, model-independent	survives as the radiation reaction
$n \geq 3$, $\propto d^n\mathbf{v}/dt^n$	R^{n-2}, vanishing	lost in the point limit

Table 1. The three groups of terms in the expansion Eq. (3), and what becomes of each as the radius of the body is taken to zero. Only the middle row survives, and the classical theory of the electron is the theory of that one row.

The first row is the divergence, and it is harmless only because of its form. Being proportional to $\dot{\mathbf{v}}$, it cannot be distinguished from mass times acceleration. It can therefore be hidden in the mass,

$$m = m_{\text{bare}} + \frac{4U_0}{3c^2}. \tag{4}$$

This is the first appearance in the subject of what would later be called mass renormalization, and the source of the first defect below.

The second row is the one that matters, and its coefficient cannot depend on how the charge is arranged. With no power of R available there is nothing left to build a coefficient from except e and c, so the combination is fixed up to a pure number. A sphere, a shell and a dumbbell all return $2e^2/3c^3$, while each returns its own coefficient for the divergent term. That asymmetry is the reason the radiation-reaction term is believed and the mass term is renormalized away.

The third row is the casualty. Those terms are the only place where the internal structure of the electron would ever have shown itself, and the point limit destroys precisely that information.

Substituting Eq. (3) into $m_{\text{bare}}\dot{\mathbf{v}} = \mathbf{F} + \mathbf{F}_{\text{self}}$ and using Eq. (4) gives

$$\boxed{m\dot{\mathbf{v}} - \frac{2e^2}{3c^3}\ddot{\mathbf{v}} + \cdots = \mathbf{F}} \tag{AL}$$

now usually called the Abraham–Lorentz equation (M. Abraham 1905; H. A. Lorentz 1909), in which m is the observed mass and everything from the second term on the left is the self-force.

1.3.2. *The same result from energy balance*

There is a much shorter route, which avoids the charge distribution altogether and therefore never meets the divergent mass. Take the Larmor formula (J. Larmor 1897) for the radiated power, $P = (2e^2/3c^3)\,\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}$, and demand that the work done by the self-force account for the energy radiated over an interval,

$$\int_{t_1}^{t_2} \mathbf{F}_{\text{self}}\cdot\mathbf{v}\,dt = -\int_{t_1}^{t_2} \frac{2e^2}{3c^3}\,\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}\,dt. \tag{5}$$

Integrating the right-hand side by parts,

$$-\int_{t_1}^{t_2} \frac{2e^2}{3c^3}\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}\,dt = -\frac{2e^2}{3c^3}\Big[\dot{\mathbf{v}}\cdot\mathbf{v}\Big]_{t_1}^{t_2} + \int_{t_1}^{t_2} \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}}\cdot\mathbf{v}\,dt. \tag{6}$$

If the boundary term vanishes, as it does for periodic motion or whenever $\dot{\mathbf{v}}\cdot\mathbf{v}$ takes the same value at both ends, the two integrands may be compared:

$$\mathbf{F}_{\text{self}} = \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}}, \tag{7}$$

which is the self-force of Eq. (AL) in three lines. The mass never enters, so it stays where classical mechanics put it.

Two things should be said about what this shortcut costs. First, the conclusion is drawn from equality of integrals over a restricted class of intervals, not from any pointwise identity, so it establishes an average effect and not an instantaneous law. The third and higher derivatives of $\mathbf{v}$, which the direct calculation delivers in Eq. (3), are simply invisible to it. Second, and more interestingly, the term thrown away is not junk. The boundary term $(2e^2/3c^3)\,\dot{\mathbf{v}}\cdot\mathbf{v}$ is the Schott energy (G. A. Schott 1912), and the piece of the covariant reaction force whose time derivative it is will turn out to be the Schott term. Everything awkward about energy balance in Sec. 3 is already present here. The radiated power and the work done by the reaction force refuse to match instant by instant. Uniformly accelerated motion radiates while feeling no reaction at all. Both puzzles sit in the term this derivation is allowed to discard because the motion was assumed periodic.

1.3.3. *Three defects*

Three objections can be raised against Eq. (AL).

The mass is entirely electromagnetic, and comes out with the wrong coefficient. The absorbed term gives $m = 4U_0/3c^2$, where U_0 is the electrostatic self-energy, finite here because the radius is. The factor 4/3 is itself a symptom: a covariant accounting of the momentum of a bound charge distribution yields U_0/c^2, and the extra third appears only because the electromagnetic stress has been counted while whatever holds the sphere together has not. It is the same omission as before, now showing up as a number. E. Fermi (1922) and W. Wilson (1936) later showed that the coefficient becomes unity once the binding stresses are included. The deeper objection survives the repair. If inertia is electromagnetic in origin, a neutral particle should have no mass, and neutral particles have mass. Inertial mass cannot be a manifestation of charge. At most one may say that charge brings mass with it. Nor does Lorentz's derivation claim otherwise on its own terms, since the U_0 appearing in it is electromagnetic energy alone and contains no mechanical rest energy. The energy-balance derivation of Sec. 1.3.2 sidesteps the whole question, which is its chief attraction.

The equation is not covariant. Equation (AL) is written with d/dt in a particular frame. The repair is due to L. D. Landau & E. M. Lifshitz (1975) and is disarmingly short. Replace d/dt by $d/d\tau$ and guess $m\,\dot{u}^\alpha = F^\alpha + k\,\ddot{u}^\alpha$. The guess fails at once, because contracting with u_α and using $u^\alpha u_\alpha = -1$, whose first two derivatives give $u \cdot a = 0$ and $u \cdot \dot{a} = -a \cdot a$, leaves $-k\,(a \cdot a)$ on the right where zero is required. The cure is to add a term along u^α and let the constraint fix its coefficient: demanding $u_\alpha(\dot{a}^\alpha + C\,u^\alpha a \cdot a) = 0$ gives $C = -1$, so that the reaction force must be

$$\frac{2e^2}{3c^3}\left(\dot{a}^\alpha - u^\alpha a^\beta a_\beta\right) = \frac{2e^2}{3c^3}\left({\delta^\alpha}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta. \tag{8}$$

This is exactly the term that Dirac's far longer construction produces in Sec. 2. Covariance alone, applied to a nonrelativistic result, very nearly hands over the answer. This is worth keeping in mind when judging how much the tube derivation really establishes.

A free electron accelerates itself without bound. This is the serious defect, and it is not repaired anywhere in these notes. It is the subject of Sec. 1.4.

1.4. *The Abraham–Lorentz equation and its runaway solutions*

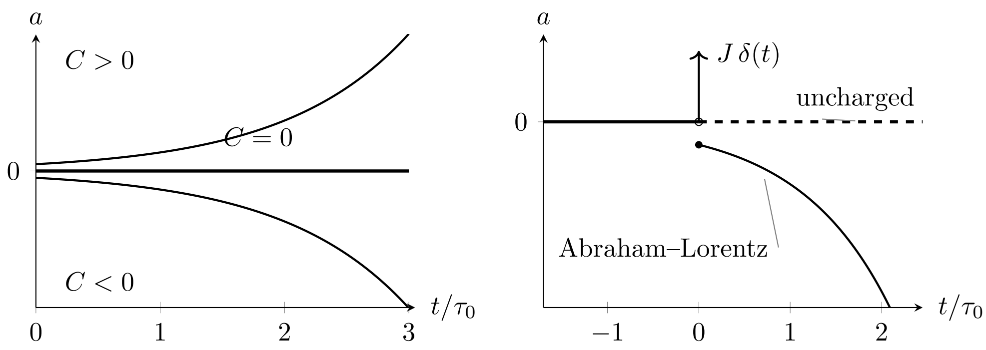
Set $\mathbf{F} = 0$ in Eq. (AL), keep only the two terms written, and reduce to one dimension. The coefficient of the self-force term is m times a time,

$$\tau_0 = \frac{2e^2}{3mc^3} = 6.27 \times 10^{-24}\ \text{s} \quad \text{(electron)}, \tag{9}$$

which will reappear in every section below, and the equation of motion of a free electron collapses to a first-order equation for the acceleration alone,

$$\dot{a} = \frac{a}{\tau_0}, \qquad \text{whence} \qquad a = C\,e^{t/\tau_0}. \tag{10}$$

Both branches of this solution are unsatisfactory, and for different reasons. If $C \neq 0$, an electron with nothing whatever acting on it accelerates itself, and does so on the timescale τ_0, which is to say it gains a factor of e in acceleration every $\sim 10^{-23}$ seconds. In the nonrelativistic equation the



(a) free electron, $a = C\,e^{t/\tau_0}$ (b) struck by an impulse at $t = 0$

Figure 1. The two faces of the third-order term. *(a)* A free electron obeying Eq. (AL) has $a = C\,e^{t/\tau_0}$. Every $C \neq 0$ runs away; the physical solution is the single value $C = 0$, and nothing in the initial-value problem selects it, since the equation is first order in a and admits any $a(0)$. *(b)* An impulse delivered at $t = 0$. An uncharged particle takes the blow, jumps in velocity, and returns to zero acceleration: the force is over and so is the acceleration. The electron does the opposite. Its acceleration jumps to $\kappa = -J/m\tau_0$ and grows without bound long after nothing is acting on it, and, since κ carries the sign opposite to J, it runs away against the blow that started it. The velocity, by contrast, is continuous at $t = 0$ here: it is the acceleration that jumps.

velocity grows without bound. In the covariant version of Sec. 3 it approaches c instead, but the character of the solution is unchanged. No intuition about free particles survives this, and no agent is available to be held responsible for the energy.

If $C = 0$ then $a = 0$, the electron moves uniformly, and Newton's first law is recovered. One may say that the Abraham–Lorentz equation tells us that a free electron moves exactly as an uncharged body does. The physical content is unobjectionable. What is objectionable is the standing of the condition. Equation (10) is first order in a and therefore admits any initial value $a(0)$. The physical solution is picked out by setting $a(0) = 0$, and the only reason offered for doing so is that every other choice misbehaves as $t \to \infty$. A condition imposed at infinity to restrict data at $t = 0$ has no warrant in the initial-value problem it is being applied to. This remains true no matter how reasonable the surviving solution looks.

The impulse problem sharpens the objection into something concrete. Write the electron's equation with an impulsive force of strength J delivered at $t = 0$,

$$ma - m\tau_0\,\dot{a} = J\,\delta(t), \tag{11}$$

and compare with the corresponding classical problem. A free mass point struck by an impulse jumps discontinuously in velocity and has zero acceleration immediately afterwards: the blow is over, and so is the acceleration. Solving Eq. (11) with the electron moving uniformly beforehand gives instead

$$a = \begin{cases} 0, & t < 0, \\ \kappa\,e^{t/\tau_0}, & t > 0, \end{cases} \qquad \kappa = -\frac{J}{m\tau_0}. \tag{12}$$

The electron does not stop accelerating when the blow ends. It accelerates ever harder, without limit, long after nothing is acting on it. Since κ carries the opposite sign to J, it runs away in the direction opposite to the blow that started it. This is the runaway, or acceleration collapse, in its starkest form. No choice of initial data avoids it here, because the requirement of uniform motion before the pulse has already been spent.

The root of the behaviour is visible in the order of the equation. Newton's equation of motion contains the acceleration and nothing beyond it, so when the force stops the acceleration stops with it. Equation (AL) contains derivatives of the acceleration, so the state of the electron at a given instant includes how fast its acceleration is changing, and that quantity survives the force that produced it. What sustains it, physically, is the electron's own electromagnetic field. The defect is therefore not an artefact of the sphere model or of the expansion used to reach Eq. (AL). Any equation of motion that accounts for the self-field by adding a derivative of the acceleration will inherit it. Dirac's equation, derived on entirely different principles in Sec. 2, inherits it exactly.

1.5. *Criteria for a physically acceptable equation of motion*

Everything from here on consists of proposals to replace Eq. (AL) with something better behaved, and proposals cannot be judged without a standard to judge them by. It is worth fixing that standard now, before any candidate is on the table, so that it cannot be quietly adjusted to fit whichever equation is under discussion. Three requirements are imposed throughout these notes.

1. **The motion must follow from the equation in the ordinary way.** The particle should obey a differential equation that can be solved by the usual procedure. One supplies initial or boundary data, integrates, and obtains the state at all later times. The data supplied must themselves be data the physical situation could actually provide. This is a weaker demand than it may appear, and Sec. 1.4 has already shown Eq. (AL) failing it: the equation is solvable, but the solution that survives is picked out by a condition imposed at $t \to \infty$, and no experimenter preparing an electron at $t = 0$ has access to that condition.

2. **A stable state must remain stable under a small disturbance.** The concrete test used throughout is the one already applied in Sec. 1.4: take a particle in uniform motion, strike it with a pulse, and ask what it does afterwards. An acceptable equation returns it to uniform motion. It is also acceptable for the particle to settle into a regular variation confined to a bounded range of speeds. This weaker alternative is included deliberately, because Sec. 5 produces solutions of exactly this kind. What is not acceptable is for the particle to accelerate without limit toward c, or to swing back and forth between $+c$ and $-c$. Both are read here as signs that the equation has no stable state to return to, rather than as physics.

3. **The ordinary physical laws must hold.** Energy conservation and causality are not negotiable, and an equation that purchases good behaviour by breaking either has not solved the problem but relocated it. This criterion does the least work in the pages that follow and is the easiest to forget, which is why it is worth writing down. The sign of a radiation-reaction term, for instance, is exactly the kind of thing one may change to remove a runaway without noticing what it does to the energy budget.

These three are the scorecard. Sections 3, 4 and 5 each close by returning to them. Two features of the list are worth noticing before it is used. It is stated entirely in terms of the behaviour of solutions and says nothing about the form an equation should take, which is deliberate, since the strategy of the sections that follow is precisely to alter the form and inspect what comes out. And it asks that a charged particle behave, in these respects, like an ordinary mechanical one. That expectation is reasonable and widely shared, but it is worth flagging as an assumption rather than a deduction.

2. THE LORENTZ–DIRAC EQUATION: DERIVATION

A word on the name before the derivation. Throughout these notes *Lorentz–Dirac*, abbreviated LD and also found in the literature as Abraham–Lorentz–Dirac, means the classical equation of motion of a point charge derived below and tagged (LD). It never means Dirac's relativistic wave equation, with which it has nothing in common but its author.

2.1. *The world-tube momentum–energy balance*

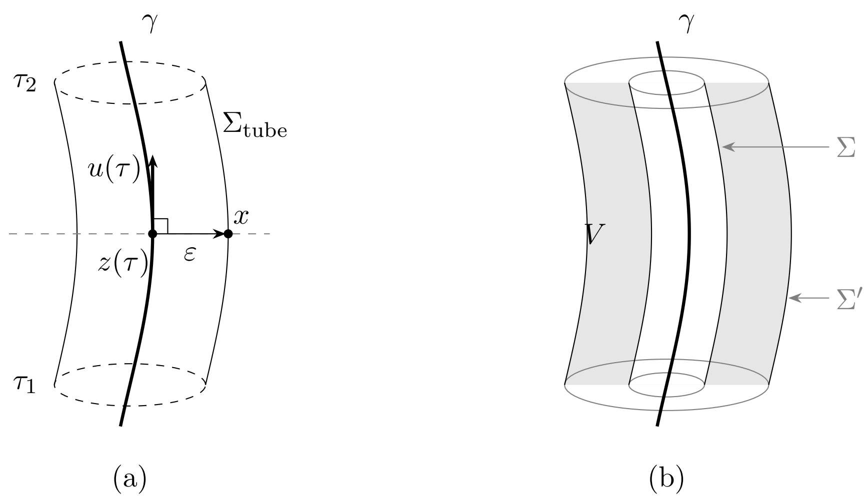


Figure 2. Dirac's construction. *(a)* The tube. At each proper time the point $z(\tau)$ on the world line γ carries a sphere of radius ε, drawn in the hyperplane of simultaneity of $z(\tau)$ — the dashed line — so that the separation $x - z(\tau)$ to a point x of the tube is orthogonal to the four-velocity $u(\tau)$, as in Eq. (15). Dragging that sphere along γ from τ_1 to τ_2 sweeps out Σ_{tube}. The end caps are drawn dashed as a reminder that the integration is carried out over the wall alone; see Sec. 2.5. *(b)* Why the choice of tube does not matter. Two tubes Σ and Σ' enclose the same world line, and the region V between them contains no charge, so the flux of $T^{\mu\nu}$ through the two is the same by Eq. (14). The balance may therefore be read on any surface, including one that keeps well away from the singular world line.

Dirac's advance was a change of question. Lorentz's route, followed in Sec. 1.3, asks what force the charge exerts on itself and answers by summing the retarded action of one charge element on another. That sum has no meaning for a point charge, since the self-field diverges on the world line. Dirac keeps the point charge and declines to ask about the force at all. He asks instead where the energy and momentum go. Whatever the particle radiates must be paid for out of the particle's own energy and momentum. That balance can be struck without ever evaluating a field on the world line. The divergent self-field still enters the calculation, but only through differences, and the differences are finite.

2.1.1. *The tube, and why its shape is immaterial*

Surround the world line γ by a three-dimensional tube Σ. The four-momentum crossing it is

$$\Delta P^{\mu} = \int_{\Sigma} T^{\mu\nu}\, d\Sigma_{\nu}. \tag{13}$$

Let Σ and Σ' both enclose γ, and let V be the region between them. That region contains no charge, so the stress–energy tensor is conserved throughout it, and

$$\Delta P'^{\mu} - \Delta P^{\mu} = \oint_{\partial V} T^{\mu\nu}\, d\Sigma_{\nu} = \int_{V} T^{\mu\nu}{}_{,\nu}\, dV = 0. \tag{14}$$

This is the whole reason the strategy beats Lorentz's. It deserves to be stated on its own. A force must be evaluated somewhere, and for a point charge every candidate location is a place where the field is infinite. A balance need not be evaluated anywhere in particular. It may be read off on any surface at all, including surfaces that keep a comfortable distance from the singularity. Equation (14) guarantees that the reading is the same. See Fig. 2(b).

Dirac's tube is the one adapted to the instantaneous rest frame. For a point x on the tube, let $z(\tau)$ be the point of γ satisfying

$$[x^{\mu} - z^{\mu}(\tau)]\, u_{\mu}(\tau) = 0, \qquad [x - z(\tau)] \cdot [x - z(\tau)] = \varepsilon^2, \tag{15}$$

the separation being spacelike. The first condition places x on the hyperplane of simultaneity of $z(\tau)$, so that "at the same moment" means at the same moment for the particle. The second fixes the radius. Together they describe a sphere of radius ε drawn in the instantaneous rest frame and dragged along the world line, sweeping out $\Sigma_{\text{tube}} = \{x(\tau, \varepsilon, \Omega);\ \tau_1 \le \tau \le \tau_2\}$, as in Fig. 2(a).

2.1.2. *The surface element*

Set $c = 1$ for the rest of this subsection and work in the instantaneous rest frame at proper time τ, restoring c at the end. Let n^{μ} be the unit spacelike vector pointing from $z(\tau)$ to the point x of the tube, so that $x = z(\tau) + \varepsilon n$ with $n \cdot u = 0$ and $n \cdot n = 1$. Three angular integrals are all that will be needed:

$$\oint d\Omega = 4\pi, \qquad \oint n^i\, d\Omega = 0, \qquad \oint n^i n^j\, d\Omega = \frac{4\pi}{3}\delta^{ij}. \tag{16}$$

Differentiate $x = z + \varepsilon n$ along the world line at fixed Ω. The vector n is Fermi–Walker transported, so $\dot{n}^{\mu} = (a \cdot n)\, u^{\mu}$. Thus

$$\frac{\partial x^{\mu}}{\partial \tau} = u^{\mu}\, (1 + \varepsilon\, a \cdot n)\, . \tag{17}$$

Each of the two angular directions contributes a factor ε. Thus

$$d\Sigma_{\nu} = n_{\nu}\, \varepsilon^2\, (1 + \varepsilon\, a \cdot n)\, d\Omega\, d\tau. \tag{18}$$

The factor $1 + \varepsilon\, a \cdot n$ is the one thing in this subsection worth memorizing. It is there because the tube is dragged along a world line that is accelerating. Successive cross-sections are not parallel, and the wall stretches on the side the particle is accelerating towards. It is also easy to write the surface element without it, and everything that distinguishes this calculation from the charged sphere of Sec. 1.3 is contained in it.

2.1.3. *The field near the world line*

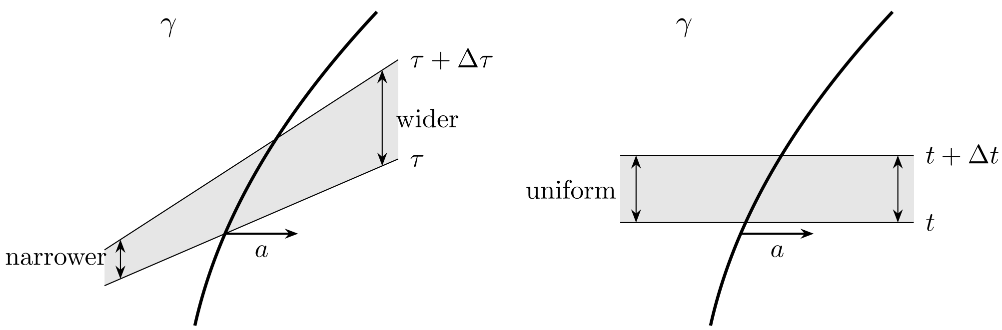
Expanded about $z(\tau)$ in the rest frame, the retarded field is

$$\mathbf{E} = \frac{e}{\varepsilon^2}\, \mathbf{n} - \frac{e}{2\varepsilon}\, [\mathbf{a} + (\mathbf{a} \cdot \mathbf{n})\mathbf{n}] + \frac{2e}{3}\, \dot{\mathbf{a}} + O(\varepsilon), \tag{19}$$

while $\mathbf{B}$, which vanishes for a charge at rest, begins only at $O(\varepsilon^{-1})$. The surface element carries ε^2, so a product of two fields must reach $O(\varepsilon^{-3})$ to leave anything divergent behind. Since $\mathbf{BB}$ is only $O(\varepsilon^{-2})$, the magnetic field cannot contribute to the divergent term at all, and the whole of the next subsubsection is an exercise with $\mathbf{E}$. The four-momentum leaving the tube is

$$\frac{dP^\mu}{d\tau} = -\oint T^{\mu\nu} n_\nu \, \varepsilon^2 \,(1 + \varepsilon\, a \cdot n)\, d\Omega, \qquad T^{\mu\nu} = \frac{1}{4\pi}\left[F^{\mu\alpha}F^{\nu}{}_{\alpha} - \tfrac{1}{4}\eta^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}\right]. \tag{20}$$

2.1.4. *The divergent term, and what becomes of the 4/3*



(a) Dirac (b) Lorentz

Figure 3. Where the 4/3 goes. Both panels show the same accelerating world line γ and the same slab of it, and differ only in what is being called "at the same moment". *(a)* Dirac's tube is built from the particle's own succession of rest frames. Because the velocity changes along γ, successive slices are not parallel: they fan open on the side the particle is accelerating towards, and the slab is wider there. That fanning is the factor $1 + \varepsilon\, a \cdot n$ of Eq. (18). *(b)* Collecting field momentum at constant time in one chosen frame gives parallel slices and a slab of uniform thickness, with no such factor. The wedge by which the two shaded regions differ is worth exactly one third of the electromagnetic mass, which is the whole of the discrepancy between Eqs. (4) and (24).

Two products of Eq. (19) reach the required order. Take the Coulomb term against itself first.

$$T^{ij} n_j \Big|_{\mathrm{CC}} = \frac{1}{4\pi}\frac{e^2}{\varepsilon^4}\left[n^i n^j - \tfrac{1}{2}\delta^{ij}\right] n^j = \frac{e^2}{8\pi\varepsilon^4}\, n^i. \tag{21}$$

At order ε^{-2} this integrates to nothing, since $\oint n^i d\Omega = 0$. An isotropic Coulomb field pushes equally in every direction. Against the tilt factor it survives:

$$-\oint \frac{e^2}{8\pi\varepsilon^4}\, n^i\, \varepsilon^3 (a \cdot n)\, d\Omega = -\frac{e^2}{8\pi\varepsilon} \cdot \frac{4\pi}{3} a^i = -\frac{e^2}{6\varepsilon}\, a^i. \tag{22}$$

The second product is the Coulomb term against the acceleration field, which is the one a sphere would have too:

$$-\oint \frac{1}{4\pi}\left[-\frac{e^2}{2\varepsilon^3} a^i - \frac{e^2}{2\varepsilon^3}(a \cdot n) n^i\right] \varepsilon^2 d\Omega = \frac{e^2}{2\varepsilon}\left(1 + \tfrac{1}{3}\right) a^i = \frac{2e^2}{3\varepsilon}\, a^i. \tag{23}$$

Writing $U_0 = e^2/2\varepsilon$ for the electrostatic energy of a shell of radius ε, the coefficient here is $4U_0/3$: the same 4/3 the sphere produced in Sec. 1.3. It is arrived at in the same way, by ignoring the tilt.

Adding Eqs. (22) and (23),

$$\left.\frac{dP^i}{d\tau}\right|_{\rm div} = \left(\frac{2}{3}-\frac{1}{6}\right)\frac{e^2}{\varepsilon}\,a^i = \frac{e^2}{2\varepsilon}\,a^i = U_0\,a^i. \tag{24}$$

The coefficient is 1, not 4/3, and the two derivations in these notes therefore disagree about the electromagnetic mass: Eq. (4) absorbed $4U_0/3c^2$, while Eq. (24) absorbs U_0/c^2. Neither contains an error, and it is worth being clear about what the disagreement is.

Section 1.3 recorded the 4/3 as the mark of having counted the electromagnetic stress while omitting whatever holds the sphere together. Dirac's calculation has no sphere to hold together and no binding stress to omit, and it returns U_0/c^2. What produces the difference is the tilt factor of Eq. (18), and the tilt factor is a statement about simultaneity. Lorentz gathers the field momentum on a hyperplane of constant time in one chosen frame. Dirac gathers it on a surface assembled from the particle's own succession of rest frames. The gap between those two ways of deciding what counts as "at the same moment" is precisely the missing third.

The 4/3 was never a statement about electrons. It was a statement about a choice of simultaneity, and it disappears as soon as the choice is made covariantly.

2.1.5. *The finite term, and two checks*

The cross of the Coulomb field with the radiative term $\frac{2e}{3}\dot{\mathbf{a}}$ gives, after the first and third pieces of $T^{ij}n_j$ cancel against each other,

$$-\oint \frac{1}{4\pi}\frac{2e^2}{3\varepsilon^2}\,\dot a^i\,\varepsilon^2\,d\Omega = -\frac{2e^2}{3}\,\dot a^i. \tag{25}$$

The finite pieces not written out are the acceleration field against itself, $\mathbf{B}$ against itself, and the tilt correction to Eq. (23). They contain no $\dot a$, and assemble into the term along u^μ that $u\cdot dP/d\tau = 0$ requires. Restoring c,

$$\frac{dP^\mu}{d\tau} = \frac{e^2}{2\varepsilon c^2}\,a^\mu - \frac{2e^2}{3c^3}\left(\dot a^\mu - u^\mu a^\beta a_\beta\right), \tag{26}$$

P^μ being the four-momentum carried out of the tube.

Two things can be checked without redoing anything. Contracting Eq. (26) with u_μ, and using $u\cdot u = -1$ and $u\cdot\dot a = -a\cdot a$, gives zero, as it must. And the component along u^μ is $(2e^2/3c^3)\,a\cdot a$, the Larmor power. Thus the radiated energy is present with the coefficient that Sec. 1.3.2 obtained from energy balance, by an argument that never mentioned a tube.

What Eq. (26) is not is an equation of motion. It records how much four-momentum the *field* carries out of the tube. Turning that into a statement about the particle requires saying what the particle's own four-momentum is, and nothing in the calculation supplies it. That freedom is the subject of Sec. 2.2.

2.2. *The undetermined four-vector B^μ*

Total four-momentum is conserved, so the particle's own four-momentum changes at the rate at which the external field supplies it, less the rate at which the field carries it out of the tube,

$$\frac{dP^\mu_{\rm mech}}{d\tau} = F^\mu_{\rm ext} - \frac{dP^\mu}{d\tau}, \tag{27}$$

with $dP^\mu/d\tau$ given by Eq. (26). Everything on the right is known. The left is not, because nothing computed so far says what P^μ_{mech} is.

This is the bill for the trick of Sec. 2.1.1. The tube integral was arranged to be insensitive to what lies inside it. That insensitivity is exactly why the divergent self-field never had to be evaluated. But the particle lies inside it. The construction is silent about the interior by design, and it is silent about the particle's own momentum for the same reason. So a form has to be put in by hand:

$$P^\mu_{\text{mech}} = m_0 u^\mu + B^\mu, \tag{28}$$

with B^μ undetermined.

One condition is available. Since $u \cdot u = -1$ throughout, $u \cdot a = 0$, and contracting Eq. (27) with u_μ must therefore give an identity. The Lorentz force is orthogonal to u, and $u \cdot dP/d\tau = 0$ was checked in Sec. 2.1.5, and $m_0\, u \cdot a = 0$. What remains is

$$u_\mu \frac{dB^\mu}{d\tau} = 0. \tag{29}$$

This says no more than that whatever force B^μ contributes must be orthogonal to the four-velocity, which is a requirement on any force term in a relativistic equation of motion. It is the only condition the tube construction imposes, though not the only one that has been imposed on B^μ in the literature: C. J. Eliezer (1947) requires in addition that $v_\lambda P^\mu_{\text{mech}} - v^\mu P_{\text{mech}\,\lambda}$ be a perfect differential, which is what forces the series of Sec. 5.5 to contain only even derivatives.

It is a very weak condition. One scalar equation restricts four functions, so three functions of τ remain free at every instant. Dirac's choice is the simplest solution,

$$B^\mu = k\, u^\mu, \qquad k = \text{const}, \tag{30}$$

for which $dB^\mu/d\tau = k\, a^\mu$ and Eq. (29) holds automatically. Substituting Eqs. (28) and (30) into Eq. (27) with Eq. (26),

$$\left(m_0 + k + \frac{e^2}{2\varepsilon c^2}\right) a^\mu = F^\mu_{\text{ext}} + \frac{2e^2}{3c^3}\left(\dot a^\mu - u^\mu a^\beta a_\beta\right), \tag{31}$$

and the entire content of Dirac's choice is visible in the bracket: k appears only added to m_0 and to the divergent $e^2/2\varepsilon c^2$. Choosing $B^\mu = k u^\mu$ is choosing to add nothing to the dynamics. It shifts a mass, and the mass is not observable separately in any case, since only the sum

$$m = m_0 + k + \frac{e^2}{2\varepsilon c^2} \tag{32}$$

appears. Holding m finite as $\varepsilon \to 0$ requires $m_0 + k \to -\infty$, which should be recorded plainly. The bare mass is negatively infinite, and the observed mass is the residue of a cancellation between two quantities. Neither quantity is separately meaningful.

Two things follow, and they point in opposite directions.

The first is that the derivation has not determined the equation of motion. It has determined it *given* Eq. (30), and Eq. (30) was chosen for simplicity, not derived. Any

$$B^\mu = P_0\, u^\mu + P_1\, a^\mu + P_2\, \dot a^\mu + \cdots \tag{33}$$

whose derivative is orthogonal to u is equally admissible, and each choice yields a different equation of motion. It is generally of higher order, since each extra term carries another derivative. Dirac himself put it as a choice of the simplest form rather than as a result. Section 5 takes that door seriously and walks through it.

The second is that the freedom is less mysterious than the notation makes it look, because a particular choice of B^μ is nothing but a choice of what to call the particle's momentum. Suppose one posits, instead of Eq. (30),

$$P^\mu_{\text{mech}} = m_0 u^\mu + C\,\frac{e^2}{c^3} a^\mu \tag{34}$$

and asks what C must be if the field side is to contain only the irreversible radiated momentum. In that bookkeeping the Schott piece has been moved from the field momentum into P^μ_{mech}. Contracting the source-free balance with u_μ, the mechanical side gives $u \cdot (m_0 a + Ce^2 \dot{a}/c^3) = -Ce^2\, a \cdot a/c^3$. The remaining radiative field piece contributes $-\frac{2}{3}e^2 a \cdot a/c^3$, so the two cancel only if

$$C = -\tfrac{2}{3}. \tag{35}$$

So orthogonality does fix the coefficient, once the *form* has been assumed. This is the same situation as before, wearing different clothes. What is worth noticing is the object it fixes. The extra piece of the momentum is $-\frac{2}{3}e^2 a^\mu/c^3$, which is the Schott momentum. It is the same quantity that appeared in Sec. 1.3.2 as the boundary term $(2e^2/3c^3)\,\dot{\mathbf{v}} \cdot \mathbf{v}$ that the energy-balance derivation was permitted to discard. It was discarded there by assuming periodic motion. Here it cannot be discarded, and it has to be carried as part of what the particle owns.

2.3. *The equation*

Absorbing the mass as in Eq. (32), Eq. (31) is the Lorentz–Dirac equation,

$$\boxed{m a^\alpha = F^\alpha_{\text{ext}} + \frac{2e^2}{3c^3}\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta} \tag{LD}$$

the projector $\delta^\alpha{}_\beta + u^\alpha u_\beta$ being the compact way of writing $\dot{a}^\alpha - u^\alpha a \cdot a$, and enforcing $u \cdot a = 0$ term by term.

The reaction force has two pieces and they are not the same kind of thing. The piece $\frac{2e^2}{3c^3}\dot{a}^\alpha$ is the Schott term. It is a total derivative, it changes sign with $\dot{a}$, and it is not radiation. It is momentum lent to the near field and returned. The piece $-\frac{2e^2}{3c^3}u^\alpha a \cdot a$ points along the four-velocity, never changes sign, and is exactly the Larmor power carried off, as Sec. 2.1.5 verified. Only the second is irreversible. Almost every puzzle in Sec. 3 is a puzzle about the first. The radiated power and the work done by the reaction force fail to match instant by instant. Uniform acceleration radiates while feeling no reaction at all.

It is worth stating what this derivation did and did not buy, since the same equation was very nearly written down in Sec. 1.3 by Landau and Lifshitz's two-line covariantization, Eq. (8). What is new is not the form. Covariance and $u \cdot a = 0$ fix the form almost by themselves, and the coefficient $2e^2/3c^3$ was already fixed by the nonrelativistic limit. What is new is, first, that no extended body was needed and therefore no binding stress was omitted, which is why the electromagnetic mass came out as U_0/c^2 rather than $4U_0/3c^2$. Second, the freedom left over has been made explicit rather than hidden, in the shape of Eq. (33). A derivation that tells you exactly which of its conclusions were choices is worth more than one that reaches the same equation silently.

What the derivation did not buy is any improvement in the behaviour of the solutions. Equation (LD) still contains $\dot{a}$, so it is still of third order in the position, and Sec. 1.4 already showed what that implies. Section 3 works out the consequences.

2.4. *The other route: separating the singular part*

Figure 4. What the retarded and advanced potentials are, geometrically. The light cone through a field point x meets the world line twice: once in the past, at $z(u)$, and once in the future, at $z(v)$. The first gives $A_{\rm ret}$, the second $A_{\rm adv}$. *(a)* For x well off the world line the two events are far apart, and only the retarded one is causally admissible. *(b)* As x is brought towards the world line the two roots approach each other and both approach the same event. Both potentials diverge there, at the same rate and with the same coefficient, because both are being sourced by the same piece of the same world line; that is why the singular parts cancel in the difference $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ and survive undiminished in the sum $\frac{1}{2}(A_{\rm ret} + A_{\rm adv})$. Equation (36) splits them where it does for this reason and no other.

There is a second way through, and it has the opposite character. Dirac's route declines to ask what force the charge exerts on itself and audits the momentum instead. This one asks about the force after all, but subtracts, before asking, the part of the self-field that makes the question meaningless.

The difficulty being addressed is the one stated in Sec. 1: at the electron both the potential and the field tensor are infinite, so the Lorentz force expression cannot be applied to the self-field. The proposal is to split the infinite self-field into an "irrelevant" singular part, which is discarded, and a finite "effective" part, and then to apply the Lorentz force law to the effective part alone, in the ordinary way. The discarded piece is called the singular term, and the procedure is the method of separating the singular part.

The split that works is between the advanced and retarded potentials. The advanced potential $A_{\rm adv}$ violates causality and is not a candidate for the physical field, but it is a perfectly good solution of $\Box A^\alpha = -4\pi j^\alpha$, and near the world line its singular behaviour is identical to that of $A_{\rm ret}$. In the difference the singular behaviour cancels. What is kept and what is thrown away are therefore

$$A^\alpha_{(R)} = \tfrac{1}{2}\left(A^\alpha_{\rm ret} - A^\alpha_{\rm adv}\right), \qquad A^\alpha_{(S)} = \tfrac{1}{2}\left(A^\alpha_{\rm ret} + A^\alpha_{\rm adv}\right), \tag{36}$$

the first being a difference of two solutions of the inhomogeneous equation and hence a solution of the homogeneous Maxwell equations, regular on the world line, with a finite Lorentz force $f^\alpha = e\,F^{(R)\alpha}{}_\beta u^\beta$.

That this reproduces Dirac's answer is the content of the equivalence theorem:

> The motion obtained from momentum–energy conservation is the same as the motion produced by the Lorentz force of the potential $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$.

Part of the check is already in hand. The finite term computed in Sec. 2.1.5 is precisely e times $\frac{1}{2}(F_{\rm ret} - F_{\rm adv})$ evaluated on the world line, which is why the same $2e^2/3c^3$ appeared there. The agreement of the two routes is not an accident, but neither is it obvious: one of them never evaluates a field at the particle and the other does nothing else.

Proofs are given by Z. Zhang (1979) and, in retarded coordinates, by E. Poisson (2004). The version reproduced here is deferred to Sec. 4.1, and it is worth saying why it gets a section of its own rather than a footnote. Its value is not that it confirms Eq. (LD) a third time. It is that carrying the proof out makes visible something the statement conceals: that Eq. (36) is a *choice* of which combination to discard, and that the choice was made on grounds of convenience. Section 4 relaxes it, replacing Eq. (36) by a two-parameter family, and everything in that section descends from the observation that the split is not forced.

2.5. *The end caps, and where the freedom in B^μ comes from*

One reservation about the derivation remains to be entered, and it turns out not to be independent of the freedom found in Sec. 2.2.

The region V occupied by the tube between τ_1 and τ_2 is bounded by the wall $\Sigma_{\rm tube}$ and by two spacelike end caps $C(\tau_1)$ and $C(\tau_2)$, the discs closing the tube at each end. Conservation applies to the whole boundary, not to part of it:

$$\int_{C(\tau_2)} T^{\mu\nu} d\Sigma_\nu - \int_{C(\tau_1)} T^{\mu\nu} d\Sigma_\nu + \int_{\Sigma_{\rm tube}} T^{\mu\nu} d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu}\, dV. \tag{37}$$

Both the calculation of Sec. 2.1 and Poisson's shorter version (E. Poisson 2004) evaluate the third term only. The 2006 text records this and does not pursue it.

The two cap integrals are not incidental. Each is the four-momentum of the field contained inside the tube at one end,

$$P^\mu_{\rm field}(\tau) = \int_{C(\tau)} T^{\mu\nu} d\Sigma_\nu, \tag{38}$$

so what Eq. (37) says is that the wall integral measures the change in the field momentum *inside* the tube, and not the change in the particle's momentum. Those are different quantities, and separating them requires knowing how the contents of the tube divide between the particle and its field.

Which is exactly what Sec. 2.2 had to put in by hand. If Eq. (38) could be evaluated, $P^\mu_{\rm mech}$ would follow from it rather than being posited, Eq. (28) would not be needed, and the four-vector B^μ would have no freedom left in it. The arbitrariness in B^μ is not a second defect standing alongside the missing caps. It is the missing caps, seen from the side of the equation of motion.

The caps are omitted because they diverge. On a cap the Coulomb energy density goes as e^2/r^4 against a volume element $r^2 dr$, and the momentum density, carrying one factor of $\mathbf{B} \sim 1/r$, as e^2/r^3; the two radial integrals

$$\int_0 \frac{dr}{r^2}, \qquad \int_0 \frac{dr}{r} \tag{39}$$

diverge at the lower limit, the second only logarithmically but no less fatally. The field momentum inside the tube is infinite, which is why it cannot be apportioned, which is why a form for $P^\mu_{\rm mech}$ has to be assumed.

This closes the account of Sec. 2.1 in a way worth stating plainly. The virtue of the tube construction and its cost are the same property. It works because it is insensitive to what lies inside it, and it is for that reason unable to say anything about what lies inside it. The divergence was not removed. It was moved to a place where it could be ignored, and the price of ignoring it is one undetermined four-vector.

A further consequence of the same divergence is left aside here. The stress–energy tensor of the electron's own field curves spacetime, and a quantity that diverges on the world line curves it without bound, so the particle is not in Minkowski space and general relativity has nothing to say about the region where the problem lives. The 2006 text raises this in its Sec. 4 and builds an "effective metric" proposal on it; that material is not covered in these notes.

3. THE LORENTZ–DIRAC EQUATION: DISCUSSION AND SOLUTIONS

Equation (LD) has been derived twice and its freedoms accounted for. This section asks what it says, and the answer is that it says very nearly everything the Abraham–Lorentz equation said in Sec. 1, in covariant dress. That is not a small disappointment. Dirac's construction repaired the derivation — no extended body, no omitted binding stress, no 4/3 — and left the disease untouched.

3.1. *One-dimensional reduction*

Restrict the motion to one spatial dimension. The normalization $u^\alpha u_\alpha = -1$ reads $\dot t^2 - \dot x^2 = 1$, which is solved identically by the rapidity $q(\tau)$,

$$\dot t = \cosh q, \qquad \dot x = \sinh q, \tag{40}$$

so that q carries the whole of the kinematics and the constraint is disposed of once and for all. Differentiating,

$$a^\alpha = \dot q\,(\sinh q,\ \cosh q), \qquad \dot a^\alpha = \ddot q\,(\sinh q,\ \cosh q) + \dot q^{\,2}(\cosh q,\ \sinh q). \tag{41}$$

Two things follow at once. First $a\cdot a = \dot q^{\,2}$, so the invariant acceleration is just $|\dot q|$. Second, and more usefully, the awkward term in Eq. (LD) collapses:

$$\dot a^\alpha - u^\alpha\, a\cdot a = \ddot q\,(\sinh q,\ \cosh q), \tag{42}$$

because the $\dot q^{\,2}$ pieces of $\dot a^\alpha$ and of $u^\alpha a\cdot a$ are identical and cancel. Both are proportional to the same vector $(\sinh q, \cosh q)$ that carries a^α, so the two components of Eq. (LD) are not independent: they reduce to a single scalar equation,

$$\dot q - \tau_0\,\ddot q = \frac{1}{m}\,f(\tau), \tag{43}$$

where $f = eE$ is the scalar force. In the normalization of the 2006 text, $m = e = c = 1$, one has $\tau_0 = 2/3$ and Eq. (43) is written $\dot q - \frac{2}{3}\ddot q = F^{01}$.

The reduction is worth pausing on, because it makes the structural point of Sec. 1.4 unmissable. Newton's equation in this variable would be $m\dot q = f$: rapidity rate proportional to force, no more. What Eq. (43) adds is $\ddot q$, and it adds it with a coefficient τ_0 that is fixed, not adjustable. The equation is first order in the acceleration $\dot q$, which is to say third order in the position.

3.2. *The free solution and the runaway*

Set $f = 0$. Equation (43) becomes $\ddot{q} = \dot{q}/\tau_0$, whence

$$q = C_1 e^{\tau/\tau_0} + C_2 \qquad \text{and} \qquad \dot{t} = \cosh\big(C_1 e^{\tau/\tau_0} + C_2\big)\,, \quad \dot{x} = \sinh\big(C_1 e^{\tau/\tau_0} + C_2\big)\,. \tag{44}$$

In the 2006 normalization this is $q = C_1 e^{3\tau/2} + C_2$.

If $C_1 \neq 0$ then $q \to \infty$, and since $dx/dt = \tanh q$, the velocity approaches c. A free electron, in an inertial frame, with nothing whatever acting on it, accelerates itself to the speed of light. The covariant equation has improved on Eq. (AL) only in the sense that the velocity now saturates at c rather than growing without bound; the world line is no less absurd.

If $C_1 = 0$ then q is constant, the electron moves uniformly, and Newton's first law is recovered. This is the physical solution, and the difficulty is the one already met in Sec. 1.4: nothing selects it. The three constants of a third-order equation are fixed by three pieces of data, and position and velocity supply only two. The third is $\dot{q}(0)$, the initial acceleration, which the physical situation does not determine and which the theory offers no way of measuring; and the only value of it that does not lead to disaster must be imposed from outside.

The impulse problem sharpens this exactly as it did before. With $f/m = \kappa\,\delta(\tau)$ and the electron moving uniformly beforehand, integrating Eq. (43) across the pulse and solving on either side gives, for $\tau > 0$,

$$\dot{q} = -\frac{\kappa}{\tau_0}\, e^{\tau/\tau_0}, \tag{45}$$

so the acceleration grows without bound after the blow instead of returning to zero. The problem Sec. 1.4 found in Lorentz's equation is untouched by Dirac's. The 2006 text notes that Zhang's treatment of the single and double pulse (Z. Zhang 1979) produces solutions of the same form.

3.3. *Dirac's asymptotic condition and the price paid*

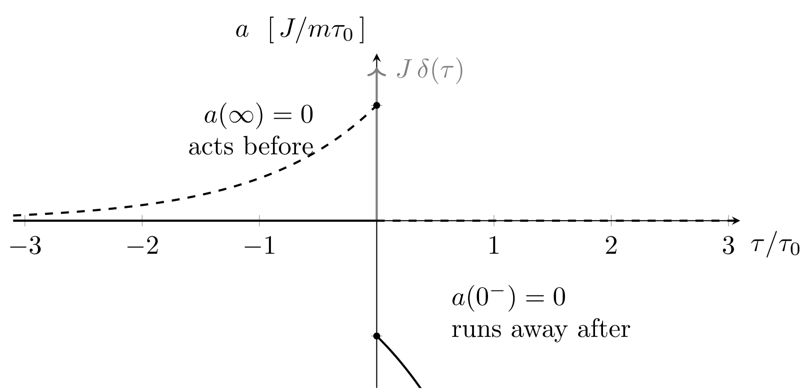


Figure 5. The choice Dirac's asymptotic condition makes. Both curves solve the same equation with the same impulse, and differ only in which end condition is imposed. Fixing the acceleration to vanish before the blow, as an initial-value problem would, gives the solid curve: nothing happens until $\tau = 0$ and the particle then runs away, as in Fig. 1(b). Fixing it to vanish at $\tau \to +\infty$ instead gives the dashed curve: the runaway is gone, the particle ends in uniform motion with exactly the velocity change J/m that the impulse should produce, and it acquires that velocity *before* the impulse arrives, over a time of order τ_0. The two branches are reflections of each other through the origin. There is no third branch, and no choice that avoids both.

Dirac's response was not to modify the equation but to change where the data are supplied. Instead of prescribing the state at some initial instant, impose

$$\dot{q}(\tau) \longrightarrow 0 \qquad \text{as} \qquad \tau \to \pm\infty, \tag{46}$$

that is, require the acceleration to die at both ends of the world line. For the impulse problem this fixes the free constant uniquely, and the solution turns out to be

$$\dot{q} = \frac{\kappa}{\tau_0}\, e^{\tau/\tau_0} \quad (\tau < 0), \qquad\qquad \dot{q} = 0 \quad (\tau > 0). \tag{47}$$

This works, and it works well. The runaway is gone. The velocity change delivered is $\int_{-\infty}^{0} \dot{q}\, d\tau = \kappa$, exactly what an impulse of that strength should produce, so the final state is the correct one. Compared with Eq. (45) the whole pathology has been reflected through $\tau = 0$, as Fig. 5 shows.

And reflecting it through $\tau = 0$ is precisely the trouble. The electron is accelerating at $\tau < 0$. It begins to move before the force reaches it, on a timescale τ_0, and it begins to move in the direction the force will push it, by the amount the force will deliver. It anticipates. Two objections follow, and the 2006 text states both.

The first is procedural. The condition $\dot{q}(0)$ is no longer required — it has been replaced by conditions at $\tau = \pm\infty$. But the motions one actually wants to describe occupy finite stretches of time, and it is a strange theory in which settling what an electron does over a microsecond requires knowing what it will be doing infinitely far in the future. The initial-value problem has not been solved; it has been exchanged for a boundary-value problem whose boundaries are not accessible.

The second is physical. Pre-acceleration is a violation of causality of a particularly naked kind: the effect is not merely correlated with the cause, it precedes it, and it precedes it by an amount that carries information about the cause. In defence one may say that $\tau_0 \sim 10^{-24}$ s is far below any timescale on which classical electrodynamics could be trusted, and that is true; but it is a statement about where the theory should be abandoned, not a resolution of the problem inside it.

What Eq. (46) buys, then, is criterion 2 of Sec. 1.5, and what it spends is criterion 3. That trade is the one fixed feature of this subject, and it will be worth remembering when Secs. 4 and 5 try to arrange a better bargain.

3.4. *Energy balance and the Schott term*

The reaction force in Eq. (LD) has two pieces, and Sec. 2.3 distinguished them by their character. Their effect on the energy budget is what makes the distinction matter. Care is needed here, because a four-force is always orthogonal to u^α. Contracting it with u_α therefore gives zero identically. It is not a power.

To speak about energy, choose an inertial observer with unit time direction n^α. The energy associated with a four-vector P^α is $E = -P \cdot n$. Write the reaction four-force as $R^\alpha = m\tau_0(\dot{a}^\alpha - u^\alpha a \cdot a)$. Its first piece is a total proper-time derivative,

$$m\tau_0\, \dot{a}^\alpha = \frac{d}{d\tau}\left(m\tau_0\, a^\alpha\right), \tag{48}$$

and the quantity being differentiated is the Schott momentum already met in Sec. 2.2. Its second piece, $-m\tau_0\, u^\alpha a\cdot a$, is the radiated four-momentum, $-P_{\rm rad} u^\alpha/c^2$ with $P_{\rm rad} = (2e^2/3c^3)\, a\cdot a$ the Larmor power in the instantaneous rest frame. The energy balance over an interval is therefore

$$\underbrace{\int_{\tau_1}^{\tau_2} -R\cdot n\,d\tau}_{\text{energy supplied by the reaction}} \quad = \quad \underbrace{\Big[-m\tau_0\,a\cdot n\Big]_{\tau_1}^{\tau_2}}_{\text{change in Schott energy}} - \underbrace{\int_{\tau_1}^{\tau_2} P_{\text{rad}}(-u\cdot n)\,d\tau}_{\text{radiated energy seen by } n}, \tag{49}$$

which is Eq. (6) again, now written with an observer specified and with the term that Sec. 1.3.2 was allowed to discard carried explicitly.

The radiated power and the work done by the reaction force therefore do not match instant by instant. They match only when the Schott term contributes nothing over the interval, which happens for periodic motion, or when the motion begins and ends with the same value of $a\cdot n$. This is the same condition, and the same evasion, as in Sec. 1.3.2. The difference is that there it was a licence to simplify a derivation, and here it is a statement about what the equation actually asserts. Between one instant and the next, the energy the particle loses and the energy the field carries away are not the same quantity. The discrepancy is stored in, and returned by, the near field.

3.5. *Hyperbolic motion*

The sharpest form of the difficulty is uniformly accelerated motion. Take

$$z^\alpha(\tau) = \frac{1}{\alpha}\left(\sinh\alpha\tau,\ \cosh\alpha\tau\right), \tag{50}$$

a hyperbola in spacetime, with constant proper acceleration α. In the rapidity variable this is simply $q=\alpha\tau$, so $\dot q=\alpha$ and $\ddot q = 0$. By Eq. (42) the reaction force vanishes identically,

$$\dot a^\alpha - u^\alpha\,a\cdot a = 0, \tag{51}$$

which is easily checked directly, since $\dot a^\alpha = \alpha^2 u^\alpha$ and $a\cdot a=\alpha^2$.

So Eq. (LD) says that a uniformly accelerated charge feels no radiation reaction at all, and follows exactly the world line an uncharged particle of the same mass would follow under the same applied force. Yet $a\cdot a=\alpha^2\neq 0$, so the Larmor formula says it radiates, steadily and forever, at the rate $(2e^2/3c^3)\alpha^2$. Something is paying for that radiation and it is not the applied force, which is doing exactly the work required to produce the motion and no more.

Equation (49) says where the accounting goes. In the inertial frame with $n^\alpha=(1,0)$, the Schott energy is $E_{\text{Schott}} = -m\tau_0 a\cdot n = m\tau_0\alpha\sinh\alpha\tau$. Its change over a finite interval is exactly $\int P_{\text{rad}}(-u\cdot n)d\tau$. The reaction force vanishes, so the left side of Eq. (49) is zero, and the radiated energy is paid for by the change in the Schott energy of the near field.

This is a consistent account, but it is not a comfortable one. The result depends on a near-field energy that grows without bound for eternal hyperbolic motion, and it ties the energy emitted in a finite interval to the history of a solution that has existed forever. The point to carry forward is not that hyperbolic motion is a contradiction. It has been discussed for a century and there are consistent ways of reading it. The point is that the trouble sits, once again, in the Schott term. Every awkward feature of Eq. (LD) traces back to the same $\dot a^\alpha$: the third order and the runaway, the pre-acceleration that removing the runaway costs, the failure of the energy budget to close instant by instant, and now a radiating particle with nothing pushing back on it.

3.6. *Scorecard, and the fork in the road*

Against the three criteria of Sec. 1.5:

1. **Solvable as an ordinary initial-value problem?** No. The equation is third order, so it needs three pieces of data, and the physical situation supplies two. The third is the initial acceleration, and every value of it but one leads to a runaway.

2. **Stable under a small disturbance?** Not as an initial-value problem: Eq. (45) runs away. Yes, if Eq. (46) is imposed instead, but then the problem is no longer an ordinary initial-value problem.

3. **Conservation and causality?** Causality fails as soon as Eq. (46) is imposed, by Eq. (47). Energy is conserved, but only in the integrated sense of Eq. (49), and Sec. 3.5 shows how uncomfortable that can get.

The pattern is worth stating in one line, because it is what the rest of these notes is a response to. As an initial-value problem, LD keeps the equation local and causal but is unstable. With Dirac's asymptotic condition it becomes stable, but only by becoming a boundary-value problem with pre-acceleration. No choice of boundary condition satisfies all three criteria, because the difficulty is not in the boundary conditions. It is in the $\dot{a}^\alpha$.

Two ways of attacking that term present themselves, and they are the subjects of the two sections that follow.

- *Sideways.* Keep the structure of the reaction term and open up its coefficient. Section 4 does this, arriving at it from the observation of Sec. 2.4 that the retarded–advanced split of Eq. (36) was a choice.

- *Upward.* Keep the coefficient and raise the order of the equation by enlarging B^μ beyond Dirac's simplest choice, along the lines left open by Eq. (33). Section 5 does this.

This contrast is the spine of the notes. One of the two routes will turn out to survive its own motivation and the other will not, but that is not visible yet, and both are pursued on their own terms.

4. SPLITTING OFF THE SINGULAR PART: THE λ–κ FAMILY

This is the sideways branch of the fork of Sec. 3.6: keep the order of the equation and open up the coefficient of the reaction term. The door was left open in Sec. 2.4. There the self-field was split into a singular part to be discarded and an effective part to be kept,

$$A^\alpha_{(R)} = \tfrac{1}{2}\left(A^\alpha_{\rm ret} - A^\alpha_{\rm adv}\right), \qquad A^\alpha_{(S)} = \tfrac{1}{2}\left(A^\alpha_{\rm ret} + A^\alpha_{\rm adv}\right),$$

the decomposition being Dirac's (P. A. M. Dirac 1938), and the modern form of the same idea being the singular–regular split of S. Detweiler & B. F. Whiting (2003). It was said in Sec. 2.4 that this split is a choice made on grounds of convenience rather than a result. This section makes good on that, by first proving that the choice reproduces Dirac's equation (Sec. 4.1), then asking whether the discarded part is really inert (Sec. 4.2), and then relaxing the split into a two-parameter family and following where the parameters lead.

Of the two branches of the fork this is the one to watch. Whatever becomes of the argument that motivates it — and Sec. 4.2 will have to be read with some care — the equation it produces is Eq. (LD) with the coefficient of the reaction term replaced by a free parameter, and that object has outlived everything else in D. Zhang (2006).

4.1. *The equivalence theorem*

> The motion obtained from momentum–energy conservation is the same as the motion produced by the Lorentz force of the potential $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$.

The theorem is not new. Proofs are given by Z. Zhang (1979) and, in retarded coordinates, by E. Poisson (2004), and neither is short. The one reproduced below is the elementary argument of D. Zhang (2006), given there precisely because the standard routes are laborious. It is worth setting out not only because it is quicker but because of what becomes visible while it is being carried out.

Start from the retarded potential of the charge (see, e.g., J. D. Jackson 1999, ch. 14),

$$A^\alpha_{\rm ret}(t, \mathbf{x}) = \int \frac{j^\alpha(t - |\mathbf{x} - \mathbf{x}'|,\ \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}\, d^3x', \tag{52}$$

and expand the source about the present time rather than the retarded one. Writing $R = |\mathbf{x} - \mathbf{x}'|$ and using $j^\alpha(t - R) = \sum_l \frac{(-R)^l}{l!} \partial_t^l j^\alpha(t)$,

$$A^\alpha_{\rm ret} = \int \sum_{l=0}^{\infty} \frac{(-1)^l}{l!}\, R^{l-1}\, \partial_t^l j^\alpha(t, \mathbf{x}')\, d^3x'. \tag{53}$$

The advanced potential is the same integral with $t + R$ in place of $t - R$, so its expansion is identical except that the factor $(-1)^l$ is absent,

$$A^\alpha_{\rm adv} = \int \sum_{l=0}^{\infty} \frac{1}{l!}\, R^{l-1}\, \partial_t^l j^\alpha(t, \mathbf{x}')\, d^3x'. \tag{54}$$

Everything now follows from the parity of l. In the half-difference the coefficient of the lth term is $\frac{1}{2}[(-1)^l - 1]$, which is zero for every even l and -1 for every odd one: *the even terms cancel and the odd terms survive.* Inserting the point-charge source, $j^0 = e\,\delta(\mathbf{x} - \mathbf{z}(t))$ and $\mathbf{j} = e\,\mathbf{v}\,\delta(\mathbf{x} - \mathbf{z}(t))$, and carrying out the $\mathbf{x}'$ integration against the delta functions,

$$\mathbf{A}_{rr} = -\sum_{l\ \rm odd} \frac{1}{l!} \frac{\partial^l}{\partial t^l}\Big[\mathbf{v}(t)\, |\mathbf{x} - \mathbf{z}(t)|^{\,l-1}\Big], \qquad \Phi_{rr} = -\sum_{l\ \rm odd} \frac{1}{l!} \frac{\partial^l}{\partial t^l}\, |\mathbf{x} - \mathbf{z}(t)|^{\,l-1}, \tag{55}$$

writing A_{rr} for $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$.

The regularity of A_{rr} can now be read off, and it is not luck. (The argument in the next two sentences is not in D. Zhang 2006, which establishes finiteness instead by noting that a difference of two solutions of the inhomogeneous equation solves the homogeneous one. The conclusion is the same; the route below makes it visible one step earlier, and generalizes.) The only term of Eqs. (53) and (54) carrying a negative power of R is $l = 0$, which is the Coulomb term; and $l = 0$ is even, so it is exactly the term the half-difference removes. Every surviving term has $l \geq 1$, hence a non-negative power R^{l-1}, and nothing in Eq. (55) blows up as $\mathbf{x} \to \mathbf{z}(t)$. This is the algebraic form of what Fig. 4 showed geometrically: the two potentials are sourced by the same piece of the same world line in the same way and differ only in the sign of the retardation, so their singular parts are identical and cancel in the difference.

The rest is computation. With A_{rr} regular one may take $\mathbf{E} = -\nabla\Phi_{rr} - \partial_t \mathbf{A}_{rr}$ and $\mathbf{B} = \nabla \times \mathbf{A}_{rr}$, evaluate them on the world line, and form $\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Carrying out the differentiations — the $l = 1$ and $l = 3$ terms of Eq. (55) are the ones that contribute — gives the three-dimensional form of the Lorentz–Dirac equation, which is the content of Appendices A.1 and A.2 of D. Zhang (2006). The theorem is proved, and the equation of Sec. 2.3 — obtained there from momentum–energy conservation with no field ever evaluated at the particle — has been recovered from a Lorentz force with a particular potential.

What the proof exposes is the thing to carry forward, and this too goes beyond D. Zhang (2006), which arrives at the same condition from the field-theoretic side rather than from the expansion. The cancellation used no property of $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ beyond the relative sign of its two pieces. For a general combination

$$A^\alpha = \lambda A^\alpha_{\rm ret} + \kappa A^\alpha_{\rm adv}, \tag{56}$$

the $l = 0$ term comes with the coefficient $\lambda + \kappa$, so

$$A^\alpha \text{ is regular on the world line} \iff \lambda + \kappa = 0. \tag{57}$$

This is the same statement as “the retained part satisfies the homogeneous Maxwell equations”, since $\lambda A^\alpha_{\rm ret} + \kappa A^\alpha_{\rm adv}$ has source $(\lambda+\kappa)j^\alpha$; the parity argument simply makes it visible one step earlier. And it says that $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ is not singled out. It is one conveniently normalized point on a line of equally regular choices, and Sec. 4.4 is what happens if one moves along that line.

4.2. *Does the singular part act?*

The split of Eq. (36) is legitimate only if the part thrown away does nothing. That is a substantive claim and D. Zhang (2006) examines it rather than assuming it. The examination is the load-bearing step of this section, and it is also the step most in need of checking, so both halves of that are set out here.

Begin with the difficulty that makes the claim non-trivial. Computing the field of $A_{(S)}$ in retarded coordinates and contracting it with the four-velocity gives a four-force whose leading behaviour is

$$f^\alpha = eF^{(S)\alpha}{}_\beta u^\beta = e^2 \left[\frac{k^\alpha - u^\alpha}{r^2} + O\left(\frac{1}{r}\right) \right], \tag{58}$$

and k^α does not approach u^α as $r \to 0$: the angle between them is fixed along each generator. So Eq. (58) is an infinite force, going as $1/r^2$, with nothing to balance it. This is the difficulty of Sec. 1 returning in its original form. To make the point picture work one has to delete an unbalanced infinity, and deleting it is not something the theory licenses.

What licenses it, in the modern treatment, is a separate principle: the singular field exerts no force on the particle, all of the self-force coming from the regular field. This is the Detweiler–Whiting principle (S. Detweiler & B. F. Whiting 2003), and D. Zhang (2006) invokes it by name, from Poisson’s review (E. Poisson 2004), while describing it frankly as a practical rule standing in for a principle that is missing.

The question is then a sharp one. *Is the Detweiler–Whiting singular field the same object as* $\frac{1}{2}(A_{\rm ret} + A_{\rm adv})$*?* If it is, the split of Eq. (36) is forced and there is nothing to relax. The 2006 text answers no, and the answer is a computation. Building the stress–energy tensor of the symmetric combination,

$$T^{\alpha\beta}_{(S)} = \frac{1}{4\pi}\left[F^{(S)\alpha}{}_{\delta}F^{\delta\beta}_{(S)} - \tfrac{1}{4}\eta^{\alpha\beta}F^{(S)}_{\mu\rho}F^{\mu\rho}_{(S)}\right], \tag{59}$$

and integrating its flux in the manner of Sec. 2.1 gives, for a unit charge, Eq. (1.2.17) of D. Zhang (2006):

$$\frac{dP^\alpha}{d\tau} = \frac{a^\alpha}{2r} + \frac{2}{3}\dot{a}^\alpha. \tag{60}$$

The first term is the familiar divergent piece proportional to the acceleration, and is absorbed into the mass as in Sec. 2.2. The second is finite and does not vanish. Read as the effect of $A_{(S)}$ on the particle, it says that the symmetric combination is *not* inert, hence is not the Detweiler–Whiting singular field, hence that Eq. (36) is one choice among others. That is the licence Sec. 4.3 uses.

Now the other half. Equation (60) should not be relied on without an independent recomputation, for two reasons.

The first is that it stands against the literature. In flat spacetime the Detweiler–Whiting singular field *is* the half-sum, and the theorem (S. Detweiler & B. F. Whiting 2003) is precisely that it exerts no force. Both statements cannot be true. The 2006 text is aware of the principle and is proposing that the identification fails; the modern position is that it does not.

The second is that the quantity computed may not be the quantity wanted. The force on the particle follows from the balance of the *total* field, and

$$T[F_{\rm ret}] = T\big[F_{(S)}\big] + T\big[F_{(R)}\big] + 2\,T\big[F_{(S)}, F_{(R)}\big], \tag{61}$$

so the flux of the singular field's own stress tensor omits the cross terms. "The momentum carried by the singular field" and "the force the singular field exerts" are different quantities, and Eq. (60) computes the first. Whether the $\frac{2}{3}\dot{a}^\alpha$ survives once Eq. (61) is treated in full, or whether it is an artefact of the algebra in Appendix A.3 of D. Zhang (2006), is not settled here.

What can be said without settling it is stated in Sec. 4.5, and it is worth flagging now so that the reader is not left waiting: the family constructed in the next two subsections does not depend on Eq. (60) for its usefulness, only for its original motivation.

4.3. *The two-parameter family*

Relax the form of the discarded part, but keep the requirement that what remains be built from the same two potentials. Writing the retained field as in Eq. (56),

$$A^\alpha = \lambda A^\alpha_{\rm ret} + \kappa A^\alpha_{\rm adv},$$

the parity decomposition of Sec. 4.1 does the work immediately. The lth term of $A_{\rm ret}$ carries $(-1)^l$ and that of $A_{\rm adv}$ carries 1, so the combination is

$$A^\alpha = (\kappa - \lambda)\sum_{l\ \rm odd}\frac{1}{l!}\,\partial_t^l[\,\cdot\,] \;+\; (\lambda + \kappa)\sum_{l\ \rm even}\frac{1}{l!}\,\partial_t^l[\,\cdot\,], \tag{62}$$

the brackets standing for the same quantities as in Eq. (55). The odd series is the regular one; the even series is the one containing $l = 0$ and hence the whole of the singularity.

Two conditions on (λ, κ) then suggest themselves, and D. Zhang (2006) takes both.

1. **Branch (a), $\lambda + \kappa = 0$.** The even series drops out entirely, so the retained field is regular on the world line. This is Eq. (57), and equivalently the statement that the retained field satisfies the homogeneous Maxwell equations. Section 4.4.
2. **Branch (b), $\lambda - \kappa = 1$.** The *discarded* part is $(1-\lambda)A^\alpha_{\text{ret}} - \kappa A^\alpha_{\text{adv}}$, and this condition makes retarded and advanced enter it with equal weight. The retained field then keeps a singular part. Section 4.5.

These are two lines in the (λ, κ) plane and they meet at one point,

$$\lambda = \tfrac{1}{2}, \qquad \kappa = -\tfrac{1}{2}, \tag{63}$$

which is Dirac's choice, Eq. (36). So the standard split is the unique member of the family satisfying both conditions at once. That is worth stating, because it explains why $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$ looks canonical: it is distinguished, just not uniquely by either requirement on its own.

A third line through the same plane is worth recording, because it was drawn first and it arrives at the same place. C. J. Eliezer (1947) takes the field of the electron to be $F_{\text{ret}} + k(F_{\text{ret}} - F_{\text{adv}})$, that is $\lambda = 1 + k$ and $\kappa = -k$, so that $\lambda + \kappa = 1$: the retained field keeps the full source rather than none of it. The singular part is then disposed of not by cancelling it but by the stipulation of Sec. 4.2 that it exerts no force, and what is left acting is $(2k+1)$ times the Dirac field. Eliezer's resulting equation of motion, his Eq. (61), is Eq. (LD) with its reaction term multiplied by $2k+1$. The two routes could hardly look less alike — one kills the singular part, the other keeps it and declares it inert — and they produce the same one-parameter family of equations.

Branch (a) can be settled here, before any equation of motion is solved. Setting $\lambda = -\kappa$ in Eq. (62) leaves $\kappa - \lambda = 2\kappa$, so

$$A^\alpha\big|_{\lambda+\kappa=0} = -2\kappa\, A^\alpha_{rr}, \tag{64}$$

with A_{rr} the Dirac field of Sec. 4.1. The retained field on this branch is nothing but Dirac's, rescaled by -2κ. Field, force and reaction term are all linear in the potential, so the reaction term of Eq. (LD) is rescaled by the same factor and nothing else changes:

$$\frac{2e^2}{3} \;\longrightarrow\; -\frac{4\kappa e^2}{3}, \tag{65}$$

which is Eq. (1.5.1) of D. Zhang (2006), recovered without solving anything. Setting $\kappa = -\frac{1}{2}$ returns the coefficient $2e^2/3$ and with it Eq. (LD), as Eq. (63) requires.

The whole of branch (a) is therefore the Lorentz–Dirac equation with one number changed. Section 4.4 has only to work out what changing it does to the solutions. Branch (b) admits no such shortcut, since the retained field keeps its singular part and the rescaling argument does not apply; it has to be handled numerically, in Sec. 4.5.

4.4. *Branch (a):* $\lambda + \kappa = 0$

Section 4.3 already established what this branch is: the Lorentz–Dirac equation with its reaction coefficient rescaled by -2κ,

$$ma^\alpha = F^\alpha_{\text{ext}} - \frac{4\kappa}{3}e^2\left(\dot{a}^\alpha - a^\beta a_\beta\, u^\alpha\right), \tag{66}$$

with $\kappa = -\frac{1}{2}$ returning Eq. (LD). What remains is to find out what the rescaling does, and for the free particle in one dimension that can be answered in closed form.

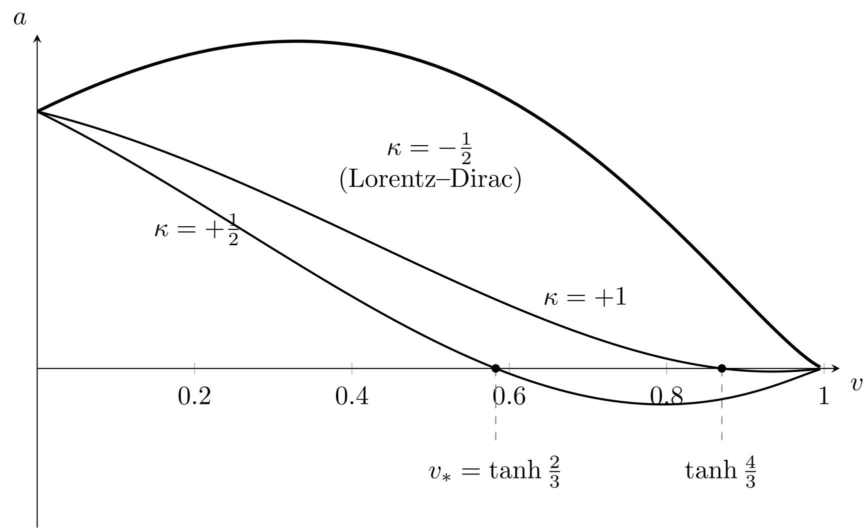


Figure 6. The phase portrait of branch (a), Eq. (68), for $C = 1$. The sign of κ decides everything. For $\kappa < 0$, which includes the Lorentz–Dirac value $\kappa = -\frac{1}{2}$, the bracket grows with v and the curve never returns to the axis: the acceleration is positive at every speed, so nothing stops the particle short of c, and this is the runaway of Sec. 3.2 seen in the (v, a) plane. For $\kappa > 0$ the bracket decreases, the curve must cross the axis, and the crossing at $v_* = \tanh(4\kappa C/3)$ is an attractor: below it the acceleration is positive and above it negative, so a particle released anywhere returns to v_*. The terminal speed is then fixed by C, which is fixed by the applied impulse — force determines velocity rather than acceleration, which is why D. Zhang (2006) calls this Aristotle motion. Only $v > 0$ is shown; the portrait is symmetric under $(v, a, C) \to (-v, -a, -C)$.

Write $a = \dot{v} = p(v)$, so that $\ddot{v} = p\, dp/dv$ and the equation of motion becomes first order in the (v, a) plane. With no external force Eq. (66) reduces to

$$\frac{dp}{dv} + \frac{3v}{1 - v^2}\, p = -\frac{3}{4\kappa}\sqrt{1 - v^2}, \tag{67}$$

a linear equation whose integrating factor is $(1 - v^2)^{-3/2}$, and which therefore integrates at once to

$$a = \left[-\frac{3}{8\kappa} \ln \frac{1 + v}{1 - v} + C\right] (1 - v^2)^{3/2}. \tag{68}$$

This is Eq. (1.2.26) of D. Zhang (2006). The constant C is fixed by the state the particle is put in — for a particle kicked from rest, by the size of the kick — and κ is the parameter of the family. Everything about this branch can be read off the one curve, and it is drawn in Fig. 6.

The sign of κ decides the whole question, and it decides it by elementary means. As v runs from 0 to 1 the logarithm runs from 0 to $+\infty$. So for $\kappa < 0$ the bracket *increases* without bound, the curve stays above the axis at every speed, and the acceleration is positive everywhere: nothing brings the particle to rest relative to its own motion and it proceeds to c. That is the runaway of Sec. 3.2, and since $\kappa = -\frac{1}{2}$ lies in this range it is also the statement that Eq. (LD) belongs to the bad half of the family.

For $\kappa > 0$ the bracket *decreases*, from C at $v = 0$ towards $-\infty$. It must therefore vanish somewhere, and where it vanishes the acceleration does too. Setting the bracket to zero and using $\ln \frac{1+v}{1-v} =$

$2\,\mathrm{artanh}\,v$,

$$v_* = \tanh\left(\frac{4\kappa C}{3}\right). \tag{69}$$

Below v_* the acceleration is positive and above it negative, so v_* is not merely a zero but an attractor: a particle released at any speed is carried to it. D. Zhang (2006) identifies the crossing and reads it as the point of re-equilibration; Eq. (69) writes it out.

That is a genuine damping. A particle disturbed from uniform motion settles back to uniform motion, which is what criterion 2 of Sec. 1.5 asks for and what nothing in Secs. 3 or 5 delivers.

It also has a peculiar character, and the 2006 text's name for it is the right one. The speed the particle ends at is set by C, and C is set by the force that was applied. *The force determines the velocity, not the acceleration.* That is Aristotle's picture of motion rather than Newton's, and Eq. (69) is its quantitative form; hence *Aristotle motion*. It is worth being clear that this is not a slip. It is what a damping term of this sign does: the reaction opposes the motion at every speed until the opposition balances the applied force, exactly as a body in a viscous medium reaches a terminal velocity, and Aristotle's mistake was to suppose that the world is always in that regime.

Two things should be entered against this branch before it is left.

The first is that κ is not determined. It is a free parameter, and the theory says only that it must be positive for the behaviour above. The 2006 text is explicit that it would have to be fixed by experiment, which is what its Sec. 5 is about and is not covered in these notes.

The second is more serious. It is not raised in the 2006 text, but it was raised in 1947, in one sentence, by C. J. Eliezer (1947): in the notation of his Eq. (61) a negative $2k+1$ turns what corresponds to self-acceleration into self-retardation, and *what corresponds to emission of radiation into absorption of radiation.* Turning κ positive reverses the sign of the reaction term in Eq. (66). The term whose sign was reversed is the one identified in Sec. 2.3 as the radiated four-momentum, $-P_{\mathrm{rad}}u^\alpha/c^2$ with $P_{\mathrm{rad}} \geq 0$ by the Larmor formula. Reversing it does not merely damp the motion; it turns radiative loss into radiative gain, and criterion 3 of Sec. 1.5 was written down with exactly this kind of move in mind. No energy budget for Eq. (66) appears anywhere in the 2006 text, and it should be computed before the $\kappa > 0$ branch is taken to have solved anything. The runaway is certainly gone. Where the energy comes from is an open question.

4.5. *Branch (b), and what the family amounts to*

Branch (b) can be dealt with quickly, because nothing later depends on it. Setting $\lambda = 1+\kappa$ leaves the retained field with a singular part, so the rescaling argument of Eq. (64) does not apply and there is no closed-form portrait; D. Zhang (2006) reduces the equation of motion and integrates it numerically with κ as the parameter. The result is one watershed and two regimes. For $\kappa > -1$, of either sign, the phase curve is convex and the electron self-accelerates much as under Eq. (LD); larger κ merely lowers the acceleration without changing its sign. At $\kappa = -1$ the portrait changes character. For $\kappa < -1$ a critical velocity v_c appears, and it is a different object from the attractor of Sec. 4.4: a particle starting below v_c remains below it for ever, approaching it asymptotically, while one starting above it decelerates stably.

One obstruction should be recorded, since it is the reason this branch is not worth pursuing further as it stands. Taking it beyond one dimension requires $\mathbf{n}\cdot\mathbf{v}$ at the position of the charge itself, where $\mathbf{n}$ is not defined. D. Zhang (2006) suggests replacing it by its angular average $\frac{1}{3}v$ and does not carry that through. Until something is done about it, branch (b) exists only in one dimension.

κ	-1.5	-2	-3	-500	-5000
v_c	0.3473	0.2261	0.1341	6.67×10^{-3}	6.67×10^{-4}

Table 2. The critical velocity of branch (b), from D. Zhang (2006). The two largest values give $|\kappa|\, v_c = 3.34$ apiece, so v_c falls off as $1/|\kappa|$ once $|\kappa|$ is large.

What the section as a whole amounts to is branch (a), and it amounts to one sentence. Equation (66) is Eq. (LD) with the coefficient of the radiation-reaction term replaced by a free parameter. Everything else — the two potentials, the parity argument, the two-parameter family, the condition $\lambda + \kappa = 0$ — is scaffolding that produces that one modification and is then dispensable.

This is worth separating carefully from the argument that motivated it. Section 4.2 rested the case for opening the split on Eq. (60), and gave reasons to doubt it: if the singular field exerts no force, as the Detweiler–Whiting principle asserts, then there was never any licence to relax Eq. (36) and the derivation collapses. But the equation it produced does not collapse with it. Written in a form that does not mention λ or κ at all,

$$ma^\alpha = F^\alpha_{\rm ext} + \zeta\, \tau_0\, m \left(\dot a^\alpha - a^\beta a_\beta\, u^\alpha\right), \qquad \zeta = -2\kappa, \tag{70}$$

with $\zeta = 1$ the standard value and $\zeta = 1$ alone recovering Eq. (LD). This is Eliezer's Eq. (61) again, with $\zeta = 2k + 1$; the parametrization is older than the derivation given for it here. This is a one-parameter deformation of the Lorentz–Dirac equation, and it stands on its own whatever becomes of the reasoning that suggested it. A derivation can be wrong and still have found something.

The virtue of Eq. (70) is that ζ is a number that data can constrain, which is more than can be said for anything else in these notes. The sign question of Sec. 4.4 is now the statement that $\zeta < 0$ reverses the radiated term, and the Aristotle branch is the whole half-line $\zeta < 0$. What the existing measurements have to say about ζ, and how tightly they bound it, is a separate matter and is not taken up here.

5. THE ELIEZER-TYPE SERIES

This is the upward branch of the fork of Sec. 3.6: keep the coefficient of the reaction term and raise the order of the equation. The door was left open in Sec. 2.2, where the tube calculation was found to fix the particle's own four-momentum only through Eq. (29), and Dirac closed it with the simplest choice.

The fork itself is not an invention of these notes, nor of D. Zhang (2006). It is stated at the head of the section C. J. Eliezer (1947) devotes to the problem: there are two ways to modify the Lorentz–Dirac equation, one being to keep the field of the electron equal to its retarded field and introduce derivatives of the velocity higher than the second, the other to abandon the purely retarded field and work with a combination of retarded and advanced. Those are Secs. 5 and 4 respectively. The 2006 text pursues both without recording that the choice between them had been posed sixty years earlier.

One warning before anything else, since two different equations carry the same name. What the literature calls the Eliezer, or Eliezer–Ford–O'Connell, equation (C. J. Eliezer 1948; G. W. Ford & R. F. O'Connell 1991) is obtained by *reducing* the order of Eq. (LD), is free of runaway solutions, and is a live candidate in present-day radiation-reaction experiments. What follows is the opposite operation, carried out on an ansatz from the earlier review (C. J. Eliezer 1947). The two sit on opposite sides of the fork just described, and reading the conclusion of this section as a verdict on the first would invert it exactly. Below, "the Eliezer equation" means the second.

5.1. *The Eliezer equation*

Return to the balance of Sec. 2.2. Written in Eliezer's notation the equation of motion before any choice is made is

$$\dot{A}_\mu - \frac{2}{3}e^2\left(\frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu\right) = e\, v_\sigma F_{\mu\,\mathrm{ext}}^{\;\sigma}, \tag{71}$$

where A_μ is the particle's own four-momentum, the P^μ_{mech} of Eq. (28). Taking $A_\mu = mv_\mu$ gives Eq. (LD); taking $A_\mu = 0$ describes a particle of zero rest mass. The question is what else A_μ may be.

Two conditions restrict it. The first is Eq. (29), $v \cdot \dot{A} = 0$. The second, which Sec. 2.2 did not impose, is that $v_\lambda A_\mu - v_\mu A_\lambda$ be a perfect differential. Together they are much stronger than either alone. Trying the obvious two-term form,

$$A_\mu = Pv_\mu + Q\dot{v}_\mu, \tag{72}$$

C. J. Eliezer (1946) showed that no choice of P and Q satisfies both: the form is simply not available. The shortest admissible extension carries one more derivative,

$$A_\mu = Pv_\mu + Q\dot{v}_\mu + R\,\frac{d\dot{v}_\mu}{d\tau}, \tag{73}$$

and imposing the two conditions on it fixes

$$P = \text{const} - \tfrac{3}{2}R\,\dot{v}^2, \qquad Q = 0, \qquad R = \text{const}. \tag{74}$$

That $Q = 0$ deserves emphasis. The coefficient of $\dot{v}_\mu$ is not small or negligible; it is forbidden. So the admissible A_μ is

$$A_\mu = mv_\mu + k\left(\frac{2}{3}\frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu\right), \tag{75}$$

with m and k constants, and substituting into Eq. (71) gives the equation this section is about,

$$\boxed{m\dot{v}_\mu - \frac{2}{3}e^2\left(\frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu\right) + k\,\frac{d}{d\tau}\left(\frac{2}{3}\frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu\right) = e\, v_\nu F_\mu^{\;\nu}} \tag{E}$$

the Eliezer equation, and Eq. (1.3.4) of D. Zhang (2006).

Set beside Eq. (LD) the difference is a single term. Everything but the k term *is* the Lorentz–Dirac equation, so Eq. (E) reduces to it at $k = 0$, and the whole of the modification is the proper-time derivative of the quantity whose undifferentiated form is already there. That costs one order: Eq. (LD) is third order in the position and Eq. (E) is fourth. It is the smallest step that can be taken along this branch, and Sec. 5.2 is what it buys.

Two remarks about the constants, both from C. J. Eliezer (1947) and neither in the 2006 text.

First, the exclusion of $\dot{v}_\mu$ is the same fact that Sec. 5.5 obtains by a different route. There the chain of identities from $v^\mu v_\mu = -1$ was used to trade odd derivatives for lower ones, leaving only even terms in the series; here the two conditions on A_μ kill the odd term directly. The two arguments meet at $Q = 0$.

Second, and more consequential, k is not free. A classical point electron has two constants, m and e, and no others, so either k describes some further property of the electron or it is expressible in terms of those two. Eliezer records that he had hoped for the first, as a way of getting Planck's constant into a classical theory, and that it did not work. Dimensionally k is a mass times a length squared, that is e^4/m, so

$$k = \sigma\,\frac{e^4}{m}, \tag{76}$$

with σ a pure number to be determined by experiment. The family therefore has one free parameter and not two, which will matter in Sec. 5.6.

5.2. *The free electron, solved*

Take Eq. (E) with no external field and in the non-relativistic limit, where it reduces to a linear equation for the three-velocity,

$$\frac{d^3\mathbf{V}}{dt^3} - 2\alpha\,\frac{d^2\mathbf{V}}{dt^2} + \beta\,\frac{d\mathbf{V}}{dt} = 0, \qquad \alpha = \frac{e^2}{2k}, \quad \beta = \frac{3m}{2k}. \tag{77}$$

This is third order, as promised, and being linear it can be solved outright. Setting $\mathbf{W} = d\mathbf{V}/dt$ leaves $\ddot{\mathbf{W}} - 2\alpha\dot{\mathbf{W}} + \beta\mathbf{W} = 0$, with roots

$$\lambda_\pm = \alpha \pm \sqrt{\alpha^2 - \beta}\,, \tag{78}$$

so that

$$\mathbf{V} = \mathbf{A} + e^{\alpha t}\left\{\mathbf{B}\,e^{+\sqrt{\alpha^2-\beta}\,t} + \mathbf{C}\,e^{-\sqrt{\alpha^2-\beta}\,t}\right\}. \tag{79}$$

Everything now depends on the signs of α and of $\alpha^2 - \beta$, and by Eq. (76) both are controlled by the single number σ:

$$\alpha = \frac{m}{2\sigma e^2}, \qquad \alpha^2 - \beta = \frac{m^2}{4\sigma^2 e^4}\,(1 - 6\sigma)\,. \tag{80}$$

Four regimes follow, and they exhaust the possibilities.

1. **$\sigma < 0$.** Then $\alpha < 0$ while $1 - 6\sigma > 1$, so $\sqrt{\alpha^2 - \beta} > |\alpha|$ and the two roots straddle zero, $\lambda_+ > 0 > \lambda_-$. If $\mathbf{B} \neq 0$ the motion is self-accelerating. If $\mathbf{B} = 0$ and $\mathbf{C} \neq 0$ it is *self-retarding*: any initial acceleration dies away and the velocity tends to a finite constant. If both vanish the motion is uniform. This is the only regime in which a physically acceptable non-trivial free motion exists.

2. **$0 < \sigma < \frac{1}{6}$.** Now $\alpha > 0$ and $0 < \sqrt{\alpha^2 - \beta} < \alpha$, so both roots are positive. Any non-zero $\mathbf{B}$ or $\mathbf{C}$ runs away, and the only physical solution is uniform motion.

3. **$\sigma = \frac{1}{6}$.** The roots coincide and $\mathbf{V} = \mathbf{A} + e^{\alpha t}(\mathbf{B}t + \mathbf{C})$, with $\alpha > 0$; again both constants must vanish.

4. **$\sigma > \frac{1}{6}$.** Here $\alpha^2 < \beta$, the roots are complex, and

$$\mathbf{V} = \mathbf{A} + e^{\alpha t}\left\{\mathbf{B}\cos\bigl[\sqrt{\beta - \alpha^2}\,t\bigr] + \mathbf{C}\sin\bigl[\sqrt{\beta - \alpha^2}\,t\bigr]\right\}. \tag{81}$$

> Since $\alpha > 0$ this is an oscillation whose amplitude grows without bound. Eliezer's comment is that such a motion is quite foreign to classical theory and cannot be compared with the wave packets of the quantum theory either.

So the free electron already settles most of the question, analytically and in 1947. Only $\sigma < 0$ admits a non-trivial acceptable motion, and even there it requires $\mathbf{B} = 0$, which is a condition on initial data of exactly the kind Sec. 3.2 objected to in Eq. (LD). Raising the order has not removed the need to impose something by hand; it has added a second constant to impose it on.

5.3. *The electron struck by a pulse, solved*

The impulse problem of Sec. 3.2 can be done in the same closed form. With a pulse of electric field $\epsilon\,\delta(t)$,

$$\frac{d^3v}{dt^3} - 2\alpha\frac{d^2v}{dt^2} + \beta\frac{dv}{dt} = \gamma\,\delta(t), \qquad \gamma = \frac{3e\epsilon}{2k}, \tag{82}$$

and for $\sigma < 0$, writing $s = \sqrt{\alpha^2 - \beta}$, the solution that stays finite at both ends is

$$v = \begin{cases} \dfrac{\gamma}{2\beta}\left(1 - \dfrac{\alpha}{s}\right) e^{(\alpha+s)t}, & t < 0, \\ \dfrac{\gamma}{\beta}\left[1 - \dfrac{1}{2}\left(1 + \dfrac{\alpha}{s}\right) e^{(\alpha-s)t}\right], & t > 0. \end{cases} \tag{83}$$

The two branches agree at $t = 0$, and the exponents have the signs found in Sec. 5.2, so $v \to 0$ as $t \to -\infty$ and $v \to \gamma/\beta$ as $t \to +\infty$.

That limit is worth evaluating. From Eq. (82),

$$\frac{\gamma}{\beta} = \frac{3e\epsilon}{2k} \cdot \frac{2k}{3m} = \frac{e\epsilon}{m}, \tag{84}$$

which is exactly the velocity a free Newtonian particle of mass m acquires from the same impulse. The equation delivers the right answer at late times, and k has dropped out of it entirely.

What k controls is the approach. The electron is already moving before the pulse arrives, on the timescale $(\alpha + s)^{-1}$, and it is still relaxing afterwards, on the timescale $|\alpha - s|^{-1}$. Eliezer's own remark is that this is *more* symmetrical than the corresponding Lorentz–Dirac solution, in which the electron accelerates until $t = 0$ and then moves uniformly. Set against Fig. 5, this is the same trade in a different currency: the runaway has been removed by a condition imposed at both ends, and the price is again that the effect precedes the cause.

5.4. *An observable that was never used*

Because Eq. (83) is explicit, so is the radiation it implies. The rate of emission is $\frac{2}{3}e^2(dv/dt)^2$, and Fourier resolving dv/dt gives the energy radiated per unit frequency,

$$E_\nu = \frac{6e^4\epsilon^2}{9\left(m - 4\pi^2k\nu^2\right)^2 + 16\pi^2e^4\nu^2}, \tag{85}$$

against the corresponding Lorentz–Dirac result, which is Eq. (85) with k set to zero. At low frequency both reduce to $\frac{2}{3}e^4\epsilon^2/m^2$, the Thomson value. At high frequency they part company: Eq. (85) falls as $\frac{2}{3}e^4\epsilon^2k^{-2}(2\pi\nu)^{-4}$, while the Lorentz–Dirac spectrum falls only as ν^{-2}.

This deserves to be flagged for what it is. It is a difference in a measurable quantity — the cross-section for scattering light off a free electron, at frequencies high enough to matter — between the Lorentz–Dirac equation and its higher-order modification, available in closed form since 1947. Every criterion in Sec. 1.5 concerns the behaviour of solutions; none of them looks at what the particle radiates. The 2006 text, which scanned parameter space numerically for signs of unphysical motion, does not mention Eq. (85), and neither did anything in Secs. 1 to 4.

One further feature is visible in Eq. (85) and is not in C. J. Eliezer (1947). For $k > 0$ the first term of the denominator vanishes at $\nu_0 = (2\pi)^{-1}\sqrt{m/k}$, so the spectrum has a resonance there, with no counterpart in the Lorentz–Dirac case. It is a sharp signature, and it belongs to the sign of k whose free-particle solutions Sec. 5.2 has already excluded.

5.5. *The general series*

Equation (75) is the shortest admissible extension, and it is where C. J. Eliezer (1947) stops: one extra derivative, one extra constant. The step to a general series is not his. It is taken in D. Zhang (2006), which writes the undetermined vector as a full series in the derivatives of the four-velocity,

$$B_\mu = P_0 v_\mu + P_1 \dot{v}_\mu + P_2 \ddot{v}_\mu + \cdots + P_i v_\mu^{(i)} + \cdots\ , \tag{86}$$

and asking which terms survive. The instrument is the normalization again. Differentiating $v^\mu v_\mu = -1$ repeatedly gives a chain,

$$\begin{aligned} v_\mu \dot{v}^\mu &= 0, \\ v_\mu \ddot{v}^\mu + \dot{v}_\mu \dot{v}^\mu &= 0, \\ v_\mu \dddot{v}^\mu + 3\dot{v}_\mu \ddot{v}^\mu &= 0, \\ v_\mu v^{(4)\mu} + 4\dot{v}_\mu \dddot{v}^\mu + 3\ddot{v}_\mu \ddot{v}^\mu &= 0, \\ v_\mu v^{(5)\mu} + 5\dot{v}_\mu v^{(4)\mu} + 10\ddot{v}_\mu \dddot{v}^\mu &= 0, \end{aligned} \tag{87}$$

whose coefficients are those of Pascal's triangle, the identities coming from the Leibniz rule; the general term is Appendix A.4 of D. Zhang (2006). Each identity lets an odd-order derivative of v_μ be traded for lower ones, so the odd terms of Eq. (86) are not independent, and what remains is

$$B_\mu = \Big\{ k - \tfrac{3}{2}P_2(\dot{v})^2 - 5P_4\, \dot{v}_\mu \dddot{v}^\mu - \tfrac{5}{2}P_4(\ddot{v})^2 - \cdots \Big\} v_\mu \ + \ P_2 \ddot{v}_\mu + P_4 v_\mu^{(4)} + \cdots + P_{2i} v_\mu^{(2i)}, \tag{88}$$

Eq. (1.3.2) of D. Zhang (2006), with one undetermined constant for each even derivative retained.

The relation to Sec. 5.1 is worth stating precisely, since the two arguments are easily confused. Eliezer's conditions kill the $\dot{v}_\mu$ term once, at the first step, and stop there; the identity chain kills every odd term at every order, and so extends the same conclusion to the whole series. Truncating Eq. (88) after P_2 returns Eq. (75), and setting every P_{2i} to zero returns Eq. (LD).

What Eq. (76) said about k applies at every order and gets worse. Each retained P_{2i} is a further constant that the theory does not fix and that no measurement has been proposed to fix, and it arrives together with two more orders of derivative.

5.6. *The relativistic equation and the 2006 scans*

The 2006 text works relativistically and numerically rather than non-relativistically and analytically. Reducing Eq. (E) to one dimension with the rapidity of Sec. 3.1 gives

$$\dddot{q} - B\ddot{q} + A\dot{q} - \tfrac{1}{2}\dot{q}^3 = 0, \qquad A = \frac{3m}{2k}, \quad B = \frac{e^2}{k}, \tag{89}$$

which is Eq. (77) with $A = \beta$, $B = 2\alpha$, plus the relativistic term $-\frac{1}{2}\dot{q}^3$. The parameters therefore carry the constraints already found: A and B have the same sign, since both are divided by k, and their ratio is fixed,

$$\frac{B}{A} = \frac{2e^2}{3m} = \tau_0 \tag{90}$$

in units with $c = 1$. There is one free number, σ, exactly as Eq. (76) says.

The scan reported in D. Zhang (2006) finds that for every sampled (A, B) the electron accelerates itself to c: monotonically for $A, B < 0$, much as under Eq. (LD), and for $A, B > 0$ with $A \gg B$ by swinging between $+c$ and $-c$ at large amplitude and high frequency. Figures 1-3-1 to 1-3-4 there show $v(\tau)$ for $(A, B) = (2, 1)$, $(3, 1)$, $(10, 1)$ and $(100, 1)$.

Those results are not independent of Sec. 5.2. The oscillatory regime of Eq. (81) is $\alpha^2 < \beta$, which in the present variables reads

$$A > \frac{B^2}{4}, \tag{91}$$

and all four sampled pairs satisfy it. The violent swinging between $\pm c$ is the relativistic form of the growing oscillation Eliezer wrote down in closed form in 1947, with the relativistic term supplying the bound at $\pm c$ that the linear equation lacks. The numerical scan rediscovered an analytic result rather than establishing a new one, and reading the two together sharpens both.

Retaining P_4 as well produces a fifth-order equation, Eq. (1.3.11) of D. Zhang (2006),

$$q^{(5)} + \left[C + 5\dot{q}^2\right]\dddot{q} - B\ddot{q} + \left[A + \tfrac{25}{2}\dot{q}^2\right]\dot{q} - \tfrac{3}{2}\dot{q}^5 - \tfrac{1}{2}C\dot{q}^3 = 0, \tag{92}$$

with $C = P_2/P_4$, whose sign is free where those of A and B are not. The scan of twenty-two triples, from rest with $q^{(4)}(0) = 2$, produces three classes of behaviour.

Behaviour	Sampled (A, B, C)
Oscillation between $\pm c$	$(100, 1, 1)$, $(100, 1, -1)$, $(100, 1, -100)$, $(1000, 1, -1000)$, $(-1, -100, 100)$
Monotonic approach to c, as under LD	$(-100, -1, 1)$, $(-100, -1, 100)$, $(-100, -1, -10)$, $(-1000, -1, -1000)$
Neither: bounded periodic pulsation	$(100, 1, 100)$, $(100, 1, 1000)$, $(1000, 1, 100)$, $(1000, 1, 1000)$, $(10000, 1, 1000)$, $(-1, -100, 1)$, $(-1, -1000, 10)$

Table 3. The fifth-order scan of D. Zhang (2006), grouped by outcome rather than in its original order. The original table lists $(100, 1, 1)$ twice with contradictory verdicts and repeats three further triples; one of each is kept here, and the class boundaries should be redone before any weight is put on them.

The first two classes are the ones already met. The third is not: the electron neither reaches c nor returns to uniform motion, but settles into a periodic pulsation of the speed within a bounded range — in the 2006 text's phrase, very like an oscillator, except that the oscillator itself keeps moving forward. This is the case criterion 2 of Sec. 1.5 was written to admit, and it is the one thing in this section that the third-order equation, and hence Eliezer's analysis, does not contain.

5.7. *What the branch was worth*

Against Sec. 1.5: criterion 1 degrades at every step, each retained P_{2i} adding two orders and two more constants that no measurement supplies; criterion 2 fails at third order for every σ except the negative ones, and there only if a constant is set to zero by hand, and is met by part of the fifth-order parameter space; criterion 3 is never tested at all.

There is also a contemporary verdict that the 2006 text does not engage. G. N. Plass (1961), reviewing the whole subject, held that the standard objections to the Lorentz–Dirac equation were largely invalid, that the extra constants of integration are fixed by requiring that the particle not gain more energy over long times than the external forces supply, and that on that prescription a physical solution exists in every case examined — including, by numerical integration, attractive and repulsive Coulomb fields, where he found no evidence of divergent or unstable solutions. On Eliezer specifically he objected that demonstrating the existence of divergent solutions proves nothing, since divergent solutions always exist, and that a non-divergent physical solution exists for each set of initial conditions. Whether or not one accepts that, it is the position the case for modifying Eq. (LD) has to answer, and neither C. J. Eliezer (1947) nor D. Zhang (2006) answers it.

There is also a sense in which nothing in this section is a finished theory, and it is worth naming precisely, because it is not the usual complaint about higher derivatives.

At Eliezer's order the equation carries one constant, k, that the theory does not determine. He is candid about having wanted it to determine something — to be a place where Planck's constant might enter a classical theory — and about that not having worked, so that k has to be written as $\sigma e^4/m$ with σ left to experiment. One undetermined number is a tolerable state of affairs; it is what Eq. (70) also amounts to, and numbers left to experiment can be measured.

The general series is a different matter. Every retained P_{2i} in Eq. (88) is another constant, the theory fixes none of them, no measurement has been proposed for any of them, and there is no principle in sight that would say where to truncate. What Eq. (86) defines is therefore not an equation of motion but a family of them indexed by an infinite sequence of unknowns — a parametrization rather than a theory. That is the respect in which the branch is incomplete, and it is incomplete by construction rather than by oversight: the two conditions on A_μ constrain the series without ever closing it.

What the branch was worth, then, is this. It produced one class of solution not available at lower order, and one observable prediction, Eq. (85), that nobody has used. It cost a free constant and two orders per step, it did not remove the need to impose a condition by hand, and it never closed. And the question the 2006 text closed on —

> Would nature really go and choose a more complicated B_μ? And if it did, we truly cannot imagine the source of nature making such a choice.

— has since been answered, though not in a way available in 2006. That answer belongs with the other twenty-years-on material and is not taken up here.

6. TWENTY YEARS ON

This section pays the debts the previous five deliberately left standing. Section 3.6 set up the fork and declined to say which branch survives. Sections 4 and 5 each followed one branch on its own terms, and each closed by saying that the verdict was not available in 2006. It is available now, and so is the answer to the question D. Zhang (2006) closed on, and so is the fate of its Sec. 5, which declared the discriminating experiment infeasible.

6.1. *Why the runaway is not a defect of the equation*

Everything in this section rests on Eq. (LD) being an effective equation rather than a fundamental one, so that claim should be established before it is used, and it is not a matter of taste. Classical electrodynamics is complete and self-consistent as a theory of fields produced by prescribed sources, and as a theory of motion in prescribed fields. It is neither of those things for a charge acting on itself. The energy in the field of a point charge,

$$\int \frac{E^2}{8\pi}\, d^3x \;\sim\; \frac{e^2}{R} \;\longrightarrow\; \infty \qquad (R \to 0), \tag{93}$$

diverges, so a point charge of finite mass is not a solution of the theory at all; and the Lorentz force eE requires the field at the particle's own location, where it is infinite, so the fundamental dynamical law cannot even be written down for the object in question. The difficulty is not that the exact equation of motion is hard to derive. **It is that classical electrodynamics contains no consistent point charge, so there is no exact equation of motion to derive: any equation of motion for a point charge is necessarily an effective equation, valid to a finite order in a small parameter, and reading it as an exact law is a category error rather than an approximation.**

Giving the charge a finite size does not rescue the programme, and the way it fails is instructive. Relativity admits no rigid bodies, so the extended charge must be held together by stresses that are not electromagnetic (H. Poincaré 1906); introducing them means leaving classical electrodynamics. Once the body has structure the answer depends on that structure, so what is obtained is the equation of motion of a particular model rather than of the electron. Only the leading term survives this arbitrariness — the coefficient τ_0 is the same for every model — which is precisely what makes the description effective, and is why the expansion of Eq. (96) below is asymptotic rather than convergent.

Two remarks keep this from sounding like a verdict of failure. The first is that the theory is never called upon to be consistent, because $r_0/\lambda_C = \alpha$: quantum electrodynamics takes over a factor $1/\alpha$ in distance before the classical self-energy becomes important, and the same α reappears in Sec. 6.3.3 as the reason the competing classical equations can never be told apart. The divergence is the theory correctly reporting the edge of its own domain. The second is that the quantum theory has the same disease in much milder form: there the self-energy grows only as $\ln \Lambda$, because the vacuum's virtual pairs smear the charge over a Compton wavelength and supply, without any Poincaré stresses, the structure the classical theory could not consistently give it. Cut both theories off at the Planck length and the classical self-energy is 10^{20} times the electron mass while the quantum one is 0.18 of it. The cure is the same operation in both cases, and Sec. 2.1.4 carried it out: absorb the divergent piece into the observed mass. P. A. M. Dirac (1938) did this nine years before the word *renormalization* existed.

With that established, the rest follows. The runaway of Sec. 3.2 is not a defect of Eq. (LD). It is what appears when every solution of an effective equation is read as a possible motion, and it disappears as soon as the equation is read as the approximation it is. That claim can be made precise using nothing beyond the one-dimensional form already derived, and this subsection does so.

Take Eq. (43) and solve it for the highest derivative rather than the lowest:

$$\tau_0\, \ddot{q} = \dot{q} - \frac{f}{m}. \tag{94}$$

The small parameter multiplies the highest derivative. That is the defining signature of a *singular* perturbation, and it carries a standard consequence: the solutions split into a low-dimensional family on which the motion is slow, and transverse directions on which it is fast.

The fast directions come out in two lines. Put $w = \dot{q}$, let $w_{\rm s}$ be any solution, and perturb, $w = w_{\rm s} + \delta$; Eq. (94) then gives $\tau_0 \dot{\delta} = \delta$, so

$$\delta = \delta_0 \, e^{\tau/\tau_0}. \tag{95}$$

This is Eq. (44) over again, and it has changed status. It is not a solution that has to be forbidden; it is the unstable transverse direction of a singularly perturbed system, and nothing physical is found on it for the reason that nothing physical is found on any unstable direction.

The slow family is equally explicit. Ask for the motions in which $\dot{q}$ is fixed by the applied force alone, and Eq. (94) is solved term by term by

$$\dot{q} = \sum_{n=0}^{\infty} \tau_0^{\,n} \frac{d^n}{d\tau^n}\left(\frac{f}{m}\right) = \frac{f}{m} + \tau_0 \frac{d}{d\tau}\left(\frac{f}{m}\right) + \tau_0^2 \frac{d^2}{d\tau^2}\left(\frac{f}{m}\right) + \cdots \,, \tag{96}$$

as substituting back confirms at once. This is the slow manifold, and it settles the question. It carries no free constant: given the force, the motion is determined by position and velocity alone, which is the Newtonian count. Equation (43) has state $(x, q, \dot{q})$, one dimension more, and that one extra dimension is Eq. (95). One extra dimension, one unstable direction, one datum that Sec. 3.2 had to supply by hand — three descriptions of the same thing.

Two complaints entered earlier can now be withdrawn in their own terms. Section 3.2 objected that $C_1 = 0$ is imposed without warrant; in the present language it says that the data lie on Eq. (96), which is an invariant set the equation itself determines and not a convention. Section 3.3's condition at $\tau = \pm\infty$ was a way of locating that same set without having the concept of it, and the pre-acceleration it introduced is the cost of the indirection rather than a property of the motion.

None of the above is an invention of these notes. The reading of Eq. (LD) as a singularly perturbed system, and of its physical solutions as a manifold rather than a selected subset, is due to H. Spohn (2000) and is set out at length in H. Spohn (2004). What is proved there, for the Abraham–Lorentz model, is that the critical manifold exists, that it is normally hyperbolic, that no solution on it runs away, and that to first order in τ_0 the motion on it obeys

$$\dot{q} = \frac{f}{m} + \tau_0 \frac{d}{d\tau}\left(\frac{f}{m}\right), \tag{97}$$

the truncation of Eq. (96) after the first correction. That equation has a name and a section of its own, and the point to carry into it is that it was not proposed as a replacement for Eq. (LD). It is Eq. (LD)'s own slow manifold, written down.

Where this stands now. The reading above is the standard one and is not in dispute; the disagreements recorded in Secs. 1 to 5, over which modification of Eq. (LD) to adopt, are closed, and they are closed because the premise they shared — that the equation needed modifying — was abandoned. What remains active is narrower and more technical. The series in Eq. (96) is asymptotic rather than convergent, and its structure and resummation have continued to be studied, as has the variational formulation of the equation and its extension to post-Newtonian settings.

6.2. *The Landau–Lifshitz equation, and why it has no free parameter*

Section 6.1 produced Eq. (97) as the first truncation of a slow manifold, in one dimension. The same reduction can be carried out directly on the covariant equation, and it is worth writing down explicitly rather than quoting, because it is a single substitution. Start from Eq. (LD),

$$ma^\alpha = F^\alpha_{\rm ext} + \tau_0\, m\,(\delta^\alpha{}_\beta + u^\alpha u_\beta)\,\dot a^\beta, \qquad \tau_0 = \frac{2e^2}{3mc^3}, \tag{98}$$

and notice that the whole correction already carries one factor of τ_0. The acceleration inside it is therefore needed only to zeroth order, and at zeroth order the equation is $ma^\alpha = F^\alpha_{\rm ext}$, so

$$\dot a^\beta = \frac{1}{m}\frac{dF^\beta_{\rm ext}}{d\tau} + O(\tau_0). \tag{99}$$

Substituting Eq. (99) into Eq. (98) gives

$$ma^\mu = F^\mu_{\rm ext} + \tau_0\left(\delta^\mu{}_\nu + c^{-2}u^\mu u_\nu\right)\frac{dF^\nu_{\rm ext}}{d\tau}, \tag{100}$$

the Landau–Lifshitz equation, given in §76 of L. D. Landau & E. M. Lifshitz (1975), with $dF/d\tau$ evaluated on the zeroth-order motion. The error committed is $O(\tau_0^2)$, which is the order Sec. 2 had already discarded in deriving Eq. (LD) in the first place, so nothing is given up that was ever there.

Its provenance is worth being exact about, because three separate results are doing the work. The equation itself is Landau and Lifshitz's. That its solutions are the physical ones — that the reduction is not merely a convenient truncation — is H. Spohn (2000). And that it follows from a controlled limit rather than from a choice is S. E. Gralla et al. (2009), discussed below. It is not boxed, and the omission is the point of Sec. 6.1: (AL), (LD) and (E) are candidate laws of motion, and Eq. (100) is not a fourth candidate but a consequence of the second.

What the substitution did, and did not do, is the point most often mistaken, and it is worth stating without hedging. **Reduction of order removes a degree of freedom, not a parameter. The coefficient τ_0 passes through Eqs. (98)–(100) untouched; what disappears is the third initial datum $a(0)$, which Sec. 3.2 had to supply by hand, and with it the runaway family that datum selects. Lorentz–Dirac and Landau–Lifshitz are therefore not two theories with different constants, nor an exact law and an approximation to it. They are one theory written before and after the solutions that were never physical have been discarded.**

Carrying out the differentiation makes the content visible. With $F^\mu_{\rm ext} = (e/c)F^{\mu\nu}u_\nu$ and $\dot u_\nu = F_\nu/m$ to the order kept,

$$\frac{dF^\mu_{\rm ext}}{d\tau} = \frac{e}{c}\,\partial_\alpha F^{\mu\nu}u^\alpha u_\nu + \frac{e^2}{mc^2}\,F^{\mu\nu}F_{\nu\lambda}u^\lambda, \tag{101}$$

so the correction consists of one term in the gradient of the external field and two quadratic in the field itself, the second of which is the radiated four-momentum. No derivative of the acceleration survives anywhere: $\dot a^\mu$ has been traded for $\dot F^\mu_{\rm ext}$, which is a property of the applied field and is known before the motion is solved for.

That trade is what buys the good behaviour. Equation (100) is second order, so it is solved from (x, v) like any equation of mechanics; there is no runaway, since Eq. (95) was the transverse direction that has been projected out; and there is no pre-acceleration, since nothing is imposed at $\tau \to \pm\infty$. All three criteria of Sec. 1.5 are met at once, which Sec. 3.6 showed Eq. (LD) cannot do under any choice of boundary condition. The price is stated plainly by Sec. 6.1: Eq. (100) does not claim to be exact, and holds to first order in τ_0.

Now the count, which is the subject of this subsection.

Equation	Free parameters	Data beyond (x, v)
Lorentz–Dirac, Eq. (LD)	0	1 per particle
Eliezer, Eq. (E)	1 (σ)	2
The series, Eq. (86)	∞	grows with order
The λ–κ family, Eq. (70)	1 (ζ)	1
Landau–Lifshitz, Eq. (100)	**0**	**0**

The coefficient in Eq. (100) is $\tau_0 = 2e^2/3mc^3$, fixed by the charge and the mass and by nothing else, and the same is true of every further term of Eq. (96). This is not the usual situation in an effective theory, where each order brings a coefficient to be measured. It happens here because a classical point charge has nothing to renormalize beyond its mass: once that is absorbed, as in Eq. (32), the expansion is determined.

The tower of Sec. 5.5 therefore collapses, and it is worth being exact about how. Different choices of B^μ correspond to adding to an effective action terms proportional to the equation of motion, and such terms are removed by a redefinition of the field variables, which changes no physical prediction. So the constants P_{2i} of Eq. (88) are not undetermined quantities awaiting measurement. They are a choice of coordinates, not new structure in the point-particle theory. Section 5.7 closed on the question of whether nature would choose a more complicated B_μ; in the point-charge limit the answer is that this is not an additional choice of physics.

Two things make that a result rather than a convention. The first is S. E. Gralla et al. (2009), who obtain the self-force from a controlled point-particle limit — a body of finite size scaled down along a one-parameter family of solutions — with no tube, no cap integrals and no undetermined four-vector. Both reservations of Sec. 2.5 are bypassed rather than answered, since the construction that produced them is never used. The second is Eq. (96) itself, which was constructed without any freedom to choose.

One qualification. All of this is for a structureless point charge. Give the particle structure and genuine parameters appear — a polarizability, a form factor — but they describe the structure, and Sec. 1.3.1 already showed where they live: the terms of Eq. (3) with $n \geq 3$, which the point limit destroys. They are not a radiation-reaction coefficient under another name.

A count of zero is a strong statement, and it invites the obvious question: if the equation has nothing to measure, what is left for an experiment to do? Section 6.3 answers it, and the answer is not that nothing is left.

Finally, a remark that belongs in these notes rather than in the literature. The author's own later drafts, `2010paper.pdf` and `2011paper.pdf` in this repository, both take Eq. (100) as their starting point and cite Poisson and Rohrlich for it. The move away from the programme of Secs. 4 and 5 was therefore made in 2010, four years after the text those sections reconstruct, and made without comment.

Table 4. The candidates, in the order they were introduced.

Equation	Order	Parameters	Status in 2026
Lorentz force	2	0	exact when radiation is negligible
Abraham–Lorentz, Eq. (AL)	3	0	teaching, non-relativistic
Lorentz–Dirac, Eq. (LD)	3	0	the reference law; not a practical evolution law
Eliezer, Eq. (E)	3	1 (σ)	not standard
The B^μ series, Eq. (86)	≥ 3	∞	field redefinitions in the point limit
λ–κ family, Eq. (70)	3	1 (ζ)	not a viable free theory; see Sec. 6.3.4
Landau–Lifshitz, Eq. (100)	2	0	**standard classical workhorse**
Eliezer–Ford–O'Connell, Eq. (89)	2	0	alternative comparison model
LL with $g(\chi)$, Eq. (107)	2	0	the quantum-corrected mean

Note—"Order" is the highest derivative of position. "Parameters" counts numbers not fixed by e and m. The kinetic descriptions of Sec. 6.4 are omitted because they are not equations of motion.

6.3. *Which equation the field uses, and why the rest cannot be told apart*

Five sections have put nine equations on the table, and two of the nine are families rather than single equations, one of them infinite. Before asking where the classical description stops (Sec. 6.4) and what has been measured (Sec. 6.5), it is worth collecting them in one place and saying what became of each. The answer is short, and it is not the answer Secs. 4 and 5 were working towards.

6.3.1. *The roster*

Table 4 is every equation these notes have written down.

The list has a shape worth noticing before any verdict is passed on it. Every entry that is third order or higher carries either a free parameter or extra initial data or both; both second-order entries carry neither. That is not a coincidence, and Sec. 6.1 explained it: an extra derivative is an extra solution, and an extra solution has to be selected by something.

6.3.2. *Each candidate was removed by a different argument, and none of them by experiment*

Four distinct arguments did the work, and it is worth keeping them separate because they are not interchangeable.

Field redefinition. The B^μ tower is not a set of undetermined constants in the point-particle theory but a relabelling of what is called the particle's position at order τ_0. Within that description there is no separate observable attached to the constants P_{2i}. This is Sec. 6.2, and it removes the infinite family as a list of independent point-charge theories.

Energy conservation. The λ–κ family varies the size of the reaction term. That size is not available to be varied: the Larmor formula follows from Maxwell's equations alone, with no equation of motion anywhere in its derivation, and requiring that the energy the particle loses equal the energy the field carries away fixes the coefficient. Eliezer made this objection in 1947 and it has not been answered since.

Spurious degrees of freedom. The higher-derivative branch of Sec. 5 adds a solution at every order. The parent theory — a body of finite size coupled to its own field — has position and velocity and nothing else, so an equation requiring $\ddot{q}$ as initial data is asking for information that does not exist. This is why the runaways of Sec. 3.2 are not a defect to be repaired but a signal that too many solutions are being carried, and why the reduction of Sec. 6.1 is the cure.

Derivation. What replaced all of it is not an argument against the alternatives at all. S. E. Gralla et al. (2009) obtain the self-force as a controlled limit of a finite body, and what comes out is the form of Eq. (100). Once the equation is derived rather than postulated there is no list to choose from.

None of these is an experimental result. Nothing in the list was excluded by data as a choice among classical point-particle forms, and Sec. 6.5 will show why: in the regime where that description applies, the differences are too small to have been isolated experimentally.

6.3.3. *Landau–Lifshitz became the standard workhorse, for three reasons*

The practical answer to "which equation" is a single one. Equation (100) is what is in the particle-in-cell codes, in the strong-field laser modelling and in much of the pulsar-magnetosphere work, and it is the classical limit against which quantum treatments are checked. Three separate things put it there.

It is the only one that can be used as an ordinary initial-value law. Equation (LD) needs an initial acceleration, and almost every choice of it runs away; a numerical integration will find the runaway whether or not the analyst wants it. Equation (100) is solved from (x, v) like any equation of mechanics. This is a mundane reason and it is probably the operative one.

Inside the classical domain nothing else is distinguishable. The candidates differ at second order in the expansion parameter $\epsilon = \tau_0\gamma\omega_B$, and Fig. 7 fixes how large that can get: the line $\epsilon = 1$ lies a factor $3/2\alpha$ above the line $\chi = 1$, so

$$\epsilon\Big|_{\chi=1} = \frac{2\alpha}{3} = 4.9\times10^{-3}, \qquad \epsilon^2\Big|_{\chi=1} = 2.4\times10^{-5}, \tag{102}$$

and at $\chi = 0.01$, where the quantum suppression of the radiated power is still only 5%, $\epsilon^2 \approx 2\times10^{-9}$. So in the whole region where the classical description is even approximately valid, the trajectories predicted by Lorentz–Dirac, Landau–Lifshitz and Eliezer–Ford–O'Connell agree to better than one part in 10^5, and where it is actually accurate they agree to one part in 10^9. *Within the classical point-particle domain, the choice among these forms is not an experimentally reachable question.* The disagreement between Landau–Lifshitz and Ford–O'Connell is real, and it is located inside Eq. (102).

It is derived, not selected. This is Sec. 6.3.2, and it is the reason the first two do not feel like a compromise.

6.3.4. *The form cannot be tested, so the coefficient is what is left*

Equation (102) has a consequence these notes have been building towards without stating it. If the differences among the candidate *forms* are 10^{-5} at the very best and 10^{-9} in practice, then eighty years of argument about which equation is correct has been an argument about something no measurement has reached in the classical point-particle domain. Everything in Secs. 1 to 5, and everything in Table 4, lives inside that gap.

What does not live inside it is the coefficient. Write

$$ma^\alpha = F^\alpha_{\text{ext}} + \zeta\, \tau_0\, m \left(\dot{a}^\alpha - a^\beta a_\beta u^\alpha\right), \qquad \zeta = 1 \text{ predicted}, \tag{103}$$

which is Eq. (70) with $\zeta = -2\kappa$, reduced in order before use. Then ζ multiplies the radiated power directly: double it and the energy loss doubles. It is the one quantity in this problem to which an experiment is linearly sensitive.

The distinction that matters is what ζ is for. It is not a free parameter of any theory on the roster — Lorentz–Dirac, Eliezer, Landau–Lifshitz and Ford–O'Connell all predict $\zeta = 1$, since all of them inherit the coefficient from Larmor. Nor is it a revival of the λ–κ family, which offered a different value as an *alternative* and was excluded for it in Sec. 6.3.2. It is a null-test parameter, introduced for the same reason γ and β are introduced in the parameterised post-Newtonian formalism: general relativity predicts $\gamma = 1$ with no freedom whatever, and one writes γ anyway, in order to say how well the prediction is known.

Said that way the gap is visible. The parameters that are free — the P_{2i}, the order of the equation — change no observable, which is why they can be infinite in number. The parameter that changes an observable is not free. These are not two statements but one, and the practical upshot is that the only number in this subject that data can constrain is a number nobody has thought to constrain, because theory fixed it and everyone believed the theory.

The state of that constraint is the subject of Sec. 6.5, and the short version is a split: the storage rings of Sec. 6.5.1 measure ζ and nothing else, at $\chi \sim 10^{-6}$, but the literature does not seem to quote a dedicated bound phrased this way; the strong-field experiments of Secs. 6.5.4 to 6.5.7 measure $g(\chi)$ while *assuming* $\zeta = 1$, so the joint constraint on the pair still appears to be missing.

6.4. *When quantum electrodynamics takes over*

Everything so far has been classical, and the question of where that stops has a sharp answer with an uncomfortable consequence: *the classical theory never reaches the field at which it would break down on its own terms.* Quantum mechanics takes over first, and by a fixed factor.

The parameter that decides is

$$\chi = \frac{\gamma B_\perp}{B_S}, \qquad B_S = \frac{m^2 c^3}{e\hbar} = 4.41 \times 10^{13}\ \text{G}, \tag{104}$$

$B_\perp$ being the field component transverse to the motion and B_S the Schwinger field. What χ measures is the fraction of the particle's own energy carried off by a single emitted photon. For $\chi \ll 1$ the electron loses its energy in many small pieces, which is what a continuous radiated power describes; for $\chi \gtrsim 1$ a single photon takes a sizeable part of it, and a description in which energy leaks out smoothly is not an approximation to that but a different thing altogether.

Three thresholds are worth separating, since they are usually run together and they fail differently. All three run the same way: the classical description requires the quantity to be *small*, and it is exceeded by going to stronger field or to higher energy, never the reverse. There is no lower boundary to the classical region.

Parameter	Classical requires	What fails once it is exceeded
$\chi = \gamma B_\perp / B_S$	$\chi \ll 1$	photon recoil is not small, and the classical emission formula is wrong at leading order, not merely corrected
$b = B/B_S$	$B \ll 4.41 \times 10^{13}$ G	Landau quantization freezes the transverse motion; there is no gyration left for an equation of motion to describe
$\tau_0 \gamma \omega_B$	$\gamma B \ll 9.07 \times 10^{15}$ G	classical radiation reaction ceases to be perturbative, and τ_0 merges with the dynamical timescale

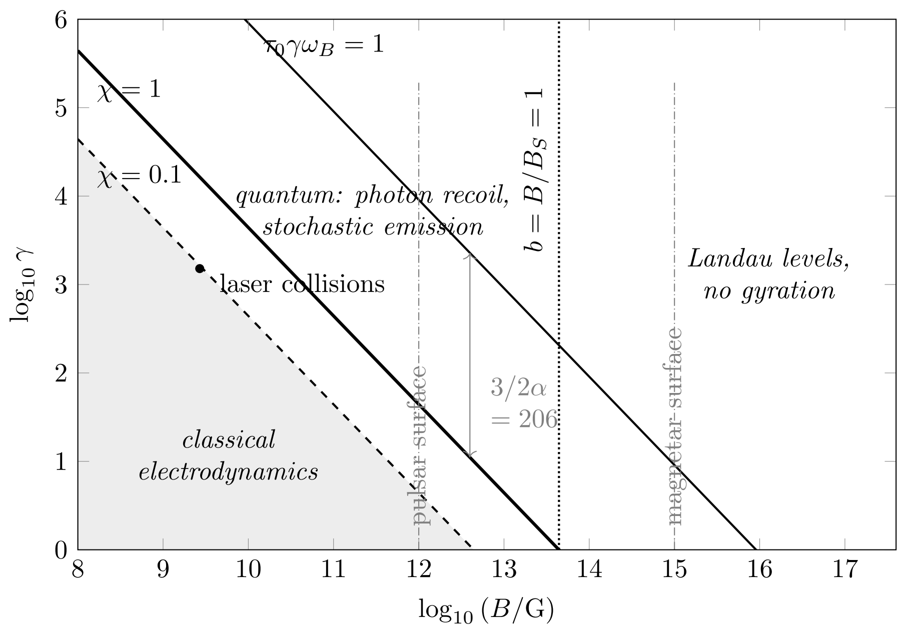


Figure 7. Where classical electrodynamics applies. Shaded is the region in which it does: $\chi \lesssim 0.1$ and $B \lesssim B_S$. Crossing *upward or to the right* across any boundary leaves it, and the three boundaries fail in different ways. Above $\chi = 1$ a single photon carries a sizeable part of the particle's energy, so the emission is quantum and stochastic. To the right of $B = B_S$ the transverse motion is Landau quantized and there is no gyration for an equation of motion to describe. The line $\tau_0\gamma\omega_B = 1$ is where classical radiation reaction would itself become non-perturbative, which is the regime Secs. 1 to 5 are about — and it lies a factor $3/2\alpha = 206$ above the line where the classical description of the emission has already failed, so it is never reached. Laser–electron collisions sit at $\chi \approx 0.09$ for $a_0 = 10$ at 800 nm against a $\gamma = 1500$ beam, the relevant field being the $2\gamma E$ seen head-on, just inside the classical region and close enough to the boundary to test it. Pulsar and magnetar surface fields are marked; at a magnetar surface even a mildly relativistic electron is already past both quantum boundaries.

Figure 7 puts the three boundaries in the (B, γ) plane. The classical region is the shaded wedge at the bottom left, and one leaves it by going up or to the right.

The third row of the table is the one Secs. 1 to 5 were about. It is also the last of the three to arrive, and the ordering is not an accident of numbers:

$$\frac{B_{\text{cl}}}{B_S} = \frac{m^2c^4/e^3}{m^2c^3/e\hbar} = \frac{\hbar c}{e^2} = \frac{1}{\alpha} \approx 137. \tag{105}$$

The same statement can be made without reference to fields at all. The classical electron radius and the reduced Compton wavelength stand in the ratio

$$\frac{r_0}{\lambda_C} = \frac{e^2/mc^2}{\hbar/mc} = \alpha, \tag{106}$$

so the length at which classical electrodynamics runs into itself lies inside the length at which quantum mechanics has already taken over, and it lies inside by exactly α. Equation (105) is Eq. (106) in the language of fields.

The consequence is worth stating flatly, because it settles the motivation of everything before this section. The runaway of Sec. 3.2, the pre-acceleration of Sec. 3.3, the merging of τ_0 with the physical timescale — these become $O(1)$ effects only at B_{cl}, and B_{cl} is 137 times past the point where the classical description of the emission has already failed. *The classical theory does not get to break down.* It is superseded well before its own pathologies become measurable, which is why no experiment has ever seen a runaway and why none ever will.

Where classical electrodynamics is still the right tool is correspondingly specific: $\chi \lesssim 0.1$. Three places qualify. Laser–electron collision experiments work deliberately at $\chi \sim 0.1$ to 1, which is the transition region and the only place the strength of the radiation reaction can be measured against a classical prediction. The outer regions of pulsar magnetospheres, far from the surface, have B low enough for classical treatment, and that is where the Aristotelian electrodynamics of Sec. 4.4 lives. And the classical result remains the $\chi \to 0$ limit that any strong-field QED calculation has to reproduce, which is how such calculations are checked.

None of this says that classical radiation reaction is wrong. Section 6.2 stands and Eq. (100) is exact in its domain; what the domain excludes is the regime in which the classical theory's own difficulties would have become visible. Nor does it say the strong-field problem is uninteresting. It says the problem there is a different one, and it is worth setting down what that problem is, because it is where the subject went.

Past $\chi \sim 1$ the classical emission formula is replaced rather than corrected. The radiated power becomes

$$P = g(\chi)\, P_{\text{cl}}, \qquad g(\chi) \simeq 0.56\, \chi^{-4/3} \quad (\chi \gg 1), \tag{107}$$

with g the quantum suppression factor, and at $\chi = 1$ the true power is already about half the classical value. That much could be absorbed into an equation of motion by rescaling a coefficient, and it is tempting to read it as the measurement of ζ that Sec. 4.5 asked for. It is not, and the reason is the second change, which no rescaling reproduces.

Emission is not continuous. The electron radiates one photon at a time, and once a single photon carries an appreciable fraction of the energy, two electrons prepared identically do not follow the same trajectory: the energy loss is a random variable and the final state is a distribution. This is straggling, and it is not a correction to a deterministic law but a statement that the law is the wrong kind of object. *No equation of motion, of any order, with any coefficient, produces a spread from* *identical initial data.* Sections 4 and 5 argued about which ordinary differential equation to write down; past $\chi \sim 1$ the answer is none of them.

What is used instead is a kinetic description: photon emission sampled event by event from the strong-field QED rates, usually under the locally-constant-field approximation, with the particle pushed classically between emissions; or a Fokker–Planck treatment when the second moment suffices; or Eq. (107) inserted into Eq. (100) when only the mean matters and the spread does not. The locally-constant-field approximation is itself the subject of current work, since it is known to fail for the low-energy part of the emitted spectrum.

6.5. *What has actually been measured*

Sections 4 and 5 argued on the only grounds available to them, whether the solutions of a proposed equation look physical. That is not the only ground available, and it is worth setting out what has been measured, because the experimental record is both older and broader than the recent laser work suggests, and most of it sits in the classical region of Fig. 7 rather than outside it.

Each subsection below is one experiment or one class of them, in order of increasing χ. Each states what the experiment establishes and gives the figure in the primary paper that carries the result, so that the claim can be checked rather than taken on trust. Table 5 collects the same information.

6.5.1. *Storage rings have been measuring the coefficient for seventy years*

Every electron storage ring is a radiation-reaction experiment that runs continuously, and the quantity it measures is the coefficient of Eq. (100) and nothing else. On a circular orbit of radius ρ the Larmor formula integrates over one turn to an energy loss

$$U_0 \;=\; \frac{4\pi}{3}\,\frac{e^2\gamma^4}{\rho}, \tag{108}$$

which is $\frac{2}{3}e^2\dot{v}^2/c^3$ carried around the ring, so the number in front of it is the same τ_0 that Secs. 1 to 5 have been arguing about. That loss is what damps the betatron and synchrotron oscillations, with damping times

$$\tau_i \;=\; \frac{2E_0T_0}{J_i\,U_0}, \qquad J_x + J_y + J_s \;=\; 4, \tag{109}$$

T_0 being the revolution period and J_i the partition numbers. The sum rule is Robinson's theorem (K. W. Robinson 1958): the damping can be moved between the three degrees of freedom by changing the lattice, but the total is fixed by the radiated power alone. Both parts of Eq. (109) are measured in every machine that is commissioned. For a 3 GeV ring with $\rho \sim 10$ m the loss is ~ 0.7 MeV per turn and the damping time a few milliseconds, some thousands of turns, which is short enough that a ring that did not damp would not store beam at all.

The regime is deeply classical: $\gamma \approx 6 \times 10^3$ in a $B = 10^4$ G bending field gives $\chi \approx 1.3 \times 10^{-6}$, at the far lower left of Fig. 7. That is exactly why the result is worth stating, because it settles a question the theoretical literature sometimes leaves open. *Radiation reaction is not a conjectural effect*, and a measured damping time is a measurement of τ_0 on a real beam. The disputes of Secs. 3 to 5 are disputes about what to do with a term whose size was settled experimentally before any of them began.

Significance. Storage rings show that classical radiation reaction is not a speculative correction. In the small-χ regime they measure the radiation-reaction coefficient through beam damping, and they do so in the domain where quantum recoil is negligible.

Where to look. There is no single canonical paper or figure here, which is itself the point: it is accelerator engineering, not a result. The standard derivation of Eqs. (108) and (109) is M. Sands (1970); the theorem is K. W. Robinson (1958).

6.5.2. *The same machines have measured the granularity of the emission since the 1970s*

Those rings also measure the quantity Sec. 6.4 called the signature of the quantum regime. Emission is by photons, so the energy loss fluctuates, and the equilibrium energy spread is set by the balance of that fluctuation against the damping of Eq. (109):

$$\left(\frac{\sigma_E}{E}\right)^2 \;=\; \frac{C_q\,\gamma^2}{J_s\,\rho}, \qquad C_q \;=\; \frac{55}{32\sqrt{3}}\,\frac{\hbar}{mc} \;=\; 3.83\times 10^{-13}\ \text{m}. \tag{110}$$

The constant is proportional to $\hbar$ and to nothing else, so a beam with a measured energy spread is a measurement of the granularity: were the emission continuous, σ_E would be zero. The same shot noise sets the horizontal emittance, and the spin polarizes radiatively by the Sokolov–Ternov effect (A. A. Sokolov & I. M. Ternov 1964) to an asymptotic $8/5\sqrt{3} = 92.4\%$, first seen at ACO and VEPP-2 around 1971.

The consequence for Sec. 6.4 is that stochastic emission is not a strong-field novelty. It has been observed for half a century at $\chi \sim 10^{-6}$, where it is a small diffusion superposed on a deterministic drift and where Eq. (110) describes it completely. What changes at $\chi \sim 1$ is not that the emission becomes stochastic — it always was — but that the diffusion stops being small compared with the drift, so that the spread can no longer be carried as a correction to a deterministic trajectory.

Significance. These measurements separate two facts that are often confused. Photon emission is always discrete, even in ordinary storage rings. What is new near $\chi \sim 1$ is not discreteness itself, but the failure of the continuous-loss approximation as the diffusion becomes large.

Where to look. M. Sands (1970) again for Eq. (110); the figure worth reproducing is a polarization build-up curve from any of the e^+e^- rings, which shows the exponential approach to 92.4% on the Sokolov–Ternov time scale.

6.5.3. *A single trapped electron shows the effect is not collective, and that it depends on the boundary*

A beam is many particles, and one could ask whether the damping is a collective effect. It is not. The cyclotron motion of one electron in a Penning trap damps radiatively at

$$\gamma_{\text{damp}} \;=\; \tau_0\,\omega_c^2, \qquad \omega_c = \frac{eB}{mc}, \tag{111}$$

which is the Abraham–Lorentz damping rate of Eq. (AL) evaluated on a circular orbit and nothing more; it is the rate for the amplitude, so the cyclotron energy decays at $2\tau_0\omega_c^2$. At $B = 5$ T, $\omega_c/2\pi = 140$ GHz and Eq. (111) gives an energy lifetime of about 0.1 s, long enough to watch directly. It is measured on one particle, which is as clean a test of the coefficient as exists.

The interesting part is that the measured rate is not always the free-space one. G. Gabrielse & H. Dehmelt (1985) found the decay inhibited by more than an order of magnitude when the cyclotron frequency was tuned away from a mode of the microwave cavity formed by the trap electrodes. This is the first observation of inhibited spontaneous emission, and it makes a point that matters for these notes: the self-force is not a property of the particle. It is a property of the particle together with the field it radiates into, and changing the boundary conditions on that field changes it. Nothing in Secs. 2 to 5 could have anticipated this, because all of them work in unbounded Minkowski space.

Significance. The Penning-trap result shows radiation damping on one electron, so it is not a collective beam effect. It also shows that the self-force depends on the radiative environment, since the cavity boundary can suppress the free-space decay rate.

Where to look. G. Gabrielse & H. Dehmelt (1985), the figure giving the measured cyclotron decay rate as the trap is tuned through the cavity resonances; the suppression is read off directly against the free-space value of Eq. (111).

6.5.4. *E-144 did in 1997 the experiment the thesis said could not be done*

The first strong-field experiment predates D. Zhang (2006) by nearly a decade. At the SLAC Final Focus Test Beam a 46.6 GeV electron beam was collided with terawatt pulses of 527 nm light at $\sim 10^{18}$ W/cm^2, giving $\eta = 0.36$ in the notation of Eq. (113) — their η is a_0 — and $\Upsilon = 0.84\,\eta = 0.3$, their Υ being χ. Two results came out of it. C. Bula et al. (1996) observed nonlinear Compton scattering, an electron absorbing several laser photons in one event. D. L. Burke et al. (1997) then observed electron–positron pairs from the collision of the backscattered GeV photon with several more laser photons, 106 ± 14 positrons above background, and this was the first observation of inelastic light-by-light scattering with real photons only. The yield scaled as

$$R_{e^+} \;\propto\; \eta^{2n}, \qquad n = 5.1 \pm 0.2\,(\mathrm{stat})\,^{+0.5}_{-0.8}\,(\mathrm{syst}), \tag{112}$$

the fifth power of the intensity, which is the multiphoton threshold made visible: the process needs at least five photons and the rate reports that number back.

This is not a radiation-reaction measurement — the electrons emitted at most a few photons, so nothing recoiled appreciably — but it establishes that $\chi \sim 0.3$ was experimentally accessible in 1997, and it is the direct counterexample to the conclusion of Sec. 5 of D. Zhang (2006) that this regime was out of reach.

Significance. E-144 did not measure radiation reaction, but it proved that the strong-field parameter range was already experimentally accessible in 1997. It also established nonlinear QED processes with real photons in the laboratory, setting the scale for later radiation-reaction experiments.

Where to look. D. L. Burke et al. (1997), Fig. 4: positron yield per laser shot against η on log axes, with the power-law fit of Eq. (112) and the shaded 95% background limit. Fig. 3 of the same paper is the raw laser-on minus laser-off positron spectrum.

6.5.5. *Aligned crystals reach $\chi \sim 1$ with no laser at all*

A crystal aligned to a beam presents a channelled particle with the coherent field of an atomic string, of order 10^{11} V/cm, and in the rest frame of a 100 GeV particle that is already comparable with B_S. T. N. Wistisen et al. (2018) sent 178.2 GeV positrons along the $\langle 111 \rangle$ axis of silicon crystals 3.8 and 10.0 mm thick, reaching $\chi \lesssim 1.4$, and measured the emitted photon spectrum. In that regime each positron emits several photons carrying an appreciable fraction of its own energy, so the recoil is not a perturbation and the spectrum is sensitive to how it is treated. They compared four calculations — classical with radiation reaction, semiclassical, quantum with radiation reaction, and quantum without — and only the quantum treatment including recoil reproduced the data.

The platform matters because its systematics have nothing in common with a laser collision. The field is static in the lab, the target is a solid with a known thickness, and the number of interacting particles is not in doubt; there is no beam overlap to estimate, which is the dominant uncertainty in everything in Sec. 6.5.6. An agreement between crystal and laser data is therefore a stronger statement than either alone.

Significance. Aligned crystals reach $\chi \sim 1$ without the beam overlap problem of laser collisions. They provide an independent route to the quantum radiation-reaction regime and show directly that recoil must be treated quantum mechanically there.

Where to look. T. N. Wistisen et al. (2018), Fig. 3: background-subtracted photon power spectrum for the two thicknesses, panels (a) and (b), with the four model curves overlaid. Fig. 4 of the same paper shows the four models on their own, without the detector response folded in, which is the cleaner figure for seeing how far apart the classical and quantum predictions actually are.

6.5.6. *The 2018 laser collisions saw radiation reaction but could not separate the models*

The modern line begins with two experiments on the same laser in the same year. The controlled quantities are

$$a_0 = \frac{eE}{mc\omega}, \qquad \chi = \frac{2\gamma a_0 \hbar\omega}{mc^2}, \tag{113}$$

the factor 2γ being the boost of the laser field into the electron rest frame in a head-on collision; $a_0 = 10$ at 800 nm against a $\gamma = 1500$ beam gives $\chi \approx 0.09$, the point marked in Fig. 7.

J. M. Cole et al. (2018) collided a laser-wakefield beam above 500 MeV with a pulse at $a_0 > 10$ and measured two things at once: the electron energy after the collision and the critical energy of the γ-ray spectrum, above 30 MeV. Neither alone is conclusive, because a_0 is not known shot to shot, but the *correlation* between them is, since each model of radiation reaction predicts a different curve in the $(\varepsilon_{\text{final}}, \varepsilon_{\text{crit}})$ plane parameterised by a_0 along its length. The data preferred a quantum description. K. Poder et al. (2018) pushed to beams above 2 GeV at 4×10^{20} W/cm^2, where the electrons see about a quarter of the critical field, and lost up to 30% of their energy; comparing against four models — Lorentz force only, Landau–Lifshitz, semiclassical, and full QED — the classical Landau–Lifshitz curve overshot the loss and the two quantum treatments did not.

Neither experiment separated the quantum treatments from one another, and neither was a high-significance exclusion of the classical one. The reason was the same in both: the dominant uncertainty is the spatial and temporal overlap of two beams a few microns across, which is not a statistical error and does not improve with more shots.

Significance. The 2018 laser experiments were the first modern laser-plasma observations of radiation reaction in the transition region. They showed measurable energy loss and a preference for quantum modelling, but they also showed that beam overlap systematics could mask the distinction between candidate models.

Where to look. J. M. Cole et al. (2018), Fig. 9 of arXiv:1707.06821: $\varepsilon_{\text{crit}}$ against $\varepsilon_{\text{final}}$, four data points, with shaded bands showing what each model would produce for a_0 uniform on $[4, 20]$ — the bands overlap, which is the honest picture of the discrimination available in 2018. K. Poder et al. (2018), Fig. 4 of arXiv:1709.01861: the measured post-collision electron spectrum against the four models in panels (a)–(d), which is the single most legible model-comparison figure in this literature.

6.5.7. *The 2026 measurement separates the models, and the classical one loses*

The discrimination that 2018 could not make has now been made. E. E. Los et al. (2026) report the first observation of strong-field radiation reaction at above 5σ in which quantum effects matter, using a 609 MeV beam against a pulse of peak $a_0 = 21.4 \pm 1.8$, with a_0 between 7 and 13 inferred for the shots analysed. The method is what the 2018 work lacked: over 600 collisions and 608 null shots, so that the difference between hit and null distributions can be established statistically, followed by a Bayesian inference on the ten highest-yield shots that treats the unknown a_0 as a parameter to be marginalised rather than a systematic to be quoted. The result is a Bayes factor favouring the quantum-continuous and quantum-stochastic models over the classical one, driven by the classical model predicting too much energy loss. The two quantum models are still not separated from each other.

The parameter to note is χ: at most 0.13 at focus and 0.05 to 0.1 for the shots used. That is not deep in the quantum regime; it is the $\chi = 0.1$ line of Fig. 7. The classical description of the emission fails, measurably and at high significance, exactly where that figure says it should, and this is the strongest evidence available that the boundary drawn there is the physical one.

Significance. Los 2026 is the first high-significance separation of a classical radiation-reaction model from quantum radiation-reaction models in the laser setting. It fixes the practical boundary of the classical description near $\chi \sim 0.1$, while leaving the quantum-continuous and quantum-stochastic descriptions unresolved.

Where to look. E. E. Los et al. (2026), arXiv:2407.12071. Fig. 4: the measured post-collision electron spectrum for the highest-yield shot against the three models. Fig. 5: the Bayes factors, shot by shot and combined, which is the figure that carries the 5σ claim and the one to reproduce if only one is wanted.

6.5.8. *LUXE and E-320 are being built to scan χ through unity*

Everything above either sits well below $\chi = 1$ or, in the crystal case, reaches it in a target that cannot be tuned. Two experiments now under construction are designed to sweep through it. LUXE at DESY will collide the 16.5 GeV European XFEL beam with a laser, and E-320 at SLAC will do the same with the 10 GeV FACET-II beam; both can vary the laser intensity at fixed beam energy, so that χ becomes a knob rather than a shot-to-shot accident. That is what is needed to test $g(\chi)$ of Eq. (107) as a function rather than at a point, and to reach the regime where the locally-constant-field approximation of Sec. 6.4 can be checked against data.

Significance. LUXE and E-320 matter because they turn χ into a controlled scan rather than an inferred shot-to-shot parameter. They are the experiments that can test the quantum suppression factor and the locally-constant-field approximation as functions, not just at isolated points.

Three things follow from the list. The first is that the existence of radiation reaction, and the accuracy of its classical description where that description applies, are not open questions and have not been for seventy years; the rings of Sec. 6.5.1 settle it, on beams and, in Sec. 6.5.3, on one particle at a time. The second is that the strong-field regime was reached in 1997, not in 2018, and the 2006 assessment of what was feasible was already nine years out of date when it was written. The third is that the classical description has now been excluded, at 5σ and at $\chi \approx 0.1$, which is the first time any of the equations in these notes has been ruled out by data rather than by argument — and what was ruled out was Eq. (100), the survivor of Secs. 1 to 5, in the one place where it was ever going to fail.

Table 5. The experimental record, in order of increasing χ.

Experiment	When	χ	Measured	Fig.
Penning trap	1985	10^{-9}	cyclotron damping, one electron	—
Storage rings	1950s–	10^{-6}	damping times, Robinson rule	—
Quantum excitation	1970s–	10^{-6}	energy spread, polarization	—
E-144	1996–97	0.3	Compton, pair creation	4
Cole 2018	2018	~ 0.1	$\varepsilon_{\rm crit}$ vs. $\varepsilon_{\rm final}$	9[a]
Poder 2018	2018	~ 0.25	electron spectrum	4[a]
Wistisen 2018	2018	$\lesssim 1.4$	photon spectrum	3
Los 2026	2026	0.05–0.1	Bayes factors, $> 5\sigma$	4, 5[a]
LUXE, E-320	building	through 1	$g(\chi)$ as a function	—

Note—Full references in the text, Secs. 6.5.1 to 6.5.8. [a]Figure numbers of the preprint version, which is the version checked here. The first three rows are standing accelerator practice rather than single experiments and have no canonical figure.

6.6. *Open questions in the field, and one missing constraint*

Here "open question" means an open question in the present literature, not a loose end internal to these notes. That distinction matters because the old classical question, which equation of motion should replace Lorentz–Dirac, is not where the field now spends its effort.

The settled side can be stated first. Eq. (LD) is an effective equation and not a fundamental one, and reduction of order is its physical content rather than a numerical convenience (Sec. 6.1). The B^μ freedom is a field redefinition in the point-particle limit (Sec. 6.2). A derivation exists that needs no world-tube and no undetermined four-vector (S. E. Gralla et al. 2009). The singular part of the field exerts no force (S. Detweiler & B. F. Whiting 2003). And Eq. (100) is the standard classical workhorse, for the reasons of Sec. 6.3.3. This is the standard reading in 2026.

6.6.1. *The physical open questions are mostly on the quantum side*

What remains is a short list, and its most striking feature is where all of it sits. The central physical items are no longer questions about the classical equation of motion.

Which description holds in the transition region. At $\chi \sim 0.1$ to 1 the candidates are the quantum-continuous treatment, in which the mean radiated power is suppressed by $g(\chi)$ of Eq. (107) and the motion stays deterministic, and stochastic emission, in which identical initial data give a distribution of final states. E. E. Los et al. (2026) excluded the classical alternative but could not separate these two, because they predict the same mean and differ only in the spread, which the incoming beam's own energy spread masks. This is a live question, and it is a question about which kinetic description is right, not which equation of motion is.

The rates themselves. The locally-constant-field approximation underlying every calculation in the previous paragraph is known to fail for the low-energy end of the emitted spectrum, and $g(\chi)$ at $\chi \gg 1$ is not tested at all. LUXE and E-320 (Sec. 6.5.8) are built to address exactly this.

Whether an equation of motion returns. Above $\chi \sim 1$ a deterministic single trajectory is no longer the right object, and the description is a kinetic equation for a distribution. Whether anything recognisable as an equation of motion survives that limit, for the mean or for a suitably defined effective particle, is open. It is the closest thing in the modern subject to the question Secs. 1 to 5 were asking.

6.6.2. *The one classical question left is mathematical, not physical*

The τ_0 expansion of Eq. (96) is asymptotic, not convergent, and its resummation and Borel structure have been studied without being settled. This is a real question and it should be listed, but its status should be stated honestly. The expansion parameter is

$$\epsilon = \tau_0 \gamma \omega_B = \frac{2\alpha}{3}\chi, \tag{114}$$

so even at the edge of the classical emission formula, $\chi = 1$, the next uncontrolled order is only

$$\epsilon^2 = \left(\frac{2\alpha}{3}\right)^2 = 2.37 \times 10^{-5}. \tag{115}$$

This is the number in Eq. (102), and it bounds the whole expansion beyond first order anywhere the classical description applies at all. Nothing observable depends on the answer. The same estimate disposes of the Landau–Lifshitz versus Ford–O'Connell disagreement, which is sometimes written as though it were open: it is open, and it is unmeasurable, and those are compatible.

6.6.3. *A missing constraint: how well is the coefficient known?*

The remaining item is different from the open questions just listed. It is not a recognised programme in the radiation-reaction literature. It is a missing way of reporting existing information.

Section 6.3.4 argued that ζ of Eq. (103) is the only number in this subject to which an experiment is linearly sensitive. The question is then simple: **what is the experimental bound on ζ?** There does not appear to be a published answer.

The reason there is none is visible in Sec. 6.5, and it is a gap between two communities rather than a hard problem. Storage-ring damping data are directly sensitive to ζ, at $\chi \sim 10^{-6}$ where $g(\chi)$ differs from unity by one part in 10^5. But to accelerator physics a damping time is a machine parameter, not a test of electrodynamics. These notes have not found a standard bound quoted in this language. The strong-field experiments of Secs. 6.5.4 to 6.5.7 do treat their measurements as tests of electrodynamics, but they test the *form*: three or four precomputed model curves, and the question of which fits. Every one of them holds $\zeta = 1$ fixed while doing so, because no reason to do otherwise has ever been offered.

Two things follow, and they are worth separating because they differ in difficulty and in interest.

The first is bookkeeping: convert existing classical-regime data into an interval for ζ. The expected answer is $\zeta = 1$, and the result is a number with an error bar rather than a discovery. It is nonetheless a number that a subject eighty years old ought to be able to quote and cannot.

The second is not bookkeeping. In the transition region the observable is $P_{\mathrm{rad}} = \zeta\, g(\chi)\, P_{\mathrm{cl}}$, in which ζ and the quantum suppression enter multiplicatively. Every strong-field result quoted in Sec. 6.5 is a statement about g obtained by holding ζ fixed. Whether those conclusions survive letting both float has not been examined. The model selection of E. E. Los et al. (2026) may be partly absorbing a coefficient. It may also be that the two are cleanly separable, in which case the answer is a joint constraint on a pair rather than a ranking of three models. Or they may be degenerate, in which case a headline result rests on an assumption nobody stated. Either outcome is worth knowing, and neither requires new data.

This is not a programme to reopen the classical theory of the electron, and it should not be written as one. The classical theory is settled for the purposes of these notes, as Secs. 6.1 to 6.3 argued at length. It is a narrow question about the observability of, and the existing constraints on, one coefficient. The sharpest risk is the obvious one: ζ may decohere against collision geometry and beam-parameter systematics badly enough that no useful interval comes out. That is a result as well, and a cheaper one to obtain than to speculate about. What remains measurable is not the choice among classical equations, but the coefficient multiplying radiation reaction.

7. WHAT CAN STILL BE DONE

[OUTLINE ONLY. Section 6.6 named a constraint that is missing and stopped there. This section is the one place in these notes that computes rather than reviews, and its job is to decide whether pursuing that constraint is worth anyone's time. State at the outset that the answer may be negative — if every observable moves only as $\zeta^{1/4}$, the astrophysical route is closed and only laboratory data remain — and that a negative answer is the useful result, not a failure of the section.]

7.1. *The equation with a free coefficient, and why it must be reduced before it can be fitted*

Section 6.3.4 introduced ζ as a null-test parameter. Using it requires an equation that can be integrated, and the form in which ζ naturally appears cannot be. Written with the coefficient free and in the projector form of Eq. (98), the equation is

$$ma^\alpha = F^\alpha_{\text{ext}} + \zeta\, \tau_0\, m\, (\delta^\alpha{}_\beta + u^\alpha u_\beta)\, \dot{a}^\beta, \tag{116}$$

which is Eq. (103), and which is the λ–κ family of Sec. 4 at $\zeta = -2\kappa$.

Equation (116) cannot be fitted to data as it stands, and the obstruction is not numerical technique. It is third order, so an integration needs $a(0)$, and Sec. 6.1 identified the extra dimension as the unstable direction of Eq. (95), which here grows as $\exp(\tau/\zeta\tau_0)$. Figure 8 is the geometry: the physical solutions lie on a surface, and every direction off it is exponentially unstable on the time scale $\zeta\tau_0$.

A component of the initial acceleration lying off that surface by one part in 10^N reaches order unity after $N \ln 10\, \zeta\tau_0$. Turning this around gives the requirement on the arithmetic: to integrate Eq. (116) over a physical time T without the runaway taking over, one must carry

$$N \;=\; \frac{T}{\zeta\tau_0 \ln 10} \tag{117}$$

decimal digits. Table 6 evaluates this for the shortest time scales anyone would want to integrate over.

The table is the argument. Double precision follows Eq. (116) for 2×10^{-22} s; reaching a single optical cycle would take a hundred and eighty million digits. **The difficulty is not that the available precision is insufficient. It is that no attainable precision helps, because the requirement**

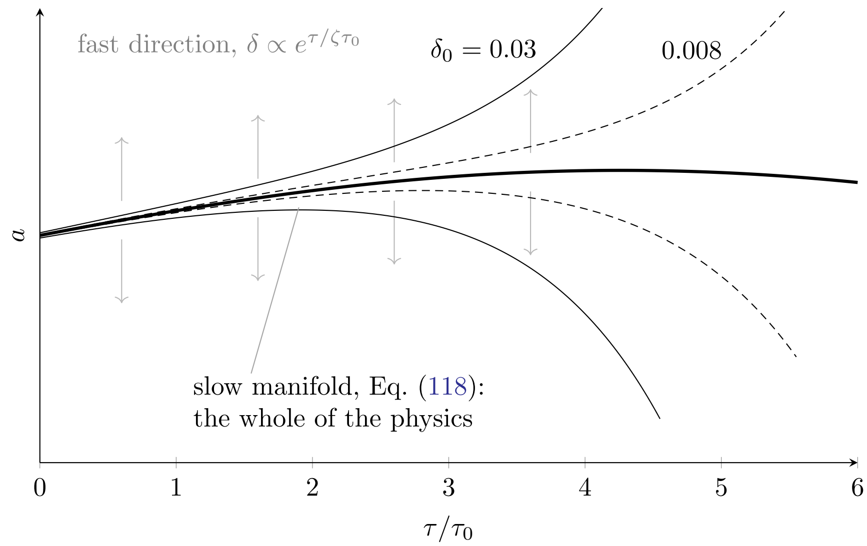


Figure 8. What reduction of order removes. The state space of the third-order equation (116) is (x, v, a); the heavy curve is the two-dimensional slow manifold on which a is fixed by the applied force through Eq. (96), and the vertical arrows are the remaining direction, along which perturbations grow as $e^{\tau/\zeta\tau_0}$. Four trajectories are shown starting off the manifold by $\delta_0 = \pm 0.03$ and ± 0.008 in units of the force; they leave the frame after $\ln(1/\delta_0)$ times $\zeta\tau_0$, and no choice of δ_0 avoids this, only postpones it logarithmically. Reduction of order is the restriction to the heavy curve, and Eq. (118) is its equation. The figure also shows why the removed object is a *coordinate* of the state space and not a coefficient: ζ sets the rate at which the thin curves depart, and moving it slides the arrows without moving the heavy curve at all.

Table 6. Decimal digits of precision required to integrate the third-order equation (116) over one physical time scale, from Eq. (117) at $\zeta = 1$.

Time scale	T (s)	T/τ_0	digits N
Laser period, 800 nm	2.7×10^{-15}	4.3×10^{8}	1.9×10^{8}
Laser pulse, 40 fs	4.0×10^{-14}	6.4×10^{9}	2.8×10^{9}
Cyclotron period, $B = 1$ T	3.6×10^{-11}	5.7×10^{12}	2.5×10^{12}
Storage-ring turn, 300 m	1.0×10^{-6}	1.6×10^{17}	6.9×10^{16}
Synchrotron cooling, $B = 1$ T, $\gamma = 10^3$	5.2×10^{-3}	8.2×10^{20}	3.6×10^{20}

Note—For comparison, double precision carries 16 digits and quadruple precision 34. Even a million-digit arithmetic would follow the equation for 1.4×10^{-17} s, seven orders of magnitude short of the first row.

grows linearly with the integration time while the arithmetic available grows not at all. What such a computation returns is the initial datum $a(0)$, amplified, and the fitted coefficient would be reporting that. **The third-order form is not a harder version of the same fit; it is a fit to a different quantity.**

The cure is the one Sec. 6.2 already applied, and it goes through unchanged. The correction in Eq. (116) carries one factor of $\zeta\tau_0$, so the acceleration inside it is needed only at zeroth order, where

Eq. (99) holds. Substituting,

$$ma^{\mu} = F^{\mu}_{\text{ext}} + \zeta\,\tau_0\left(\delta^{\mu}{}_{\nu} + c^{-2}u^{\mu}u_{\nu}\right)\frac{dF^{\nu}_{\text{ext}}}{d\tau}, \tag{118}$$

which is Eq. (100) with the coefficient carried through. This is the equation the rest of this section uses. Note what the reduction did to ζ: nothing at all, exactly as it did nothing to τ_0 in Sec. 6.2. It removed a degree of freedom and left every coefficient where it stood, which is why a free coefficient may be carried through the procedure at all.

Three things follow, and they are the reason the reduction is a precondition for the measurement rather than a convenience.

First, the radiated power is now exactly proportional to ζ,

$$P = \zeta\,P_{\text{cl}}, \qquad P_{\text{cl}} = \frac{2e^4B^2\gamma^2\beta_{\perp}^2}{3m^2c^3}, \tag{119}$$

with no admixture from the initial data, because there are no longer any initial data beyond (x, v). Every scaling in Secs. 7.2 and 7.3 descends from Eq. (119).

Second, ζ multiplies a term computed from the applied field alone. $dF_{\text{ext}}/d\tau$ is known before the trajectory is solved for, so the dependence of any observable on ζ can be tracked analytically instead of being read out of a numerical solution — which is what makes the exponents of Sec. 7.3 obtainable in closed form.

Third, a limitation that should be stated rather than discovered later. The reduction is exact only to first order in $\zeta\tau_0\gamma\omega_B$, so freeing the coefficient rescales the expansion parameter of Sec. 6.3.3 from ϵ to $\zeta\epsilon$. The construction therefore presumes ζ of order unity, and a fit that wandered to $\zeta \sim 10^5$ would have invalidated the equation it was fitting. In practice this is not a constraint worth worrying about: Eq. (102) gives $\epsilon \leq 5\times10^{-3}$ anywhere the classical description applies at all, so ζ has some four orders of magnitude of room before the expansion notices.

7.2. *The cooling rate is linear in ζ, and so is everything that balances against it*

Two consequences of Eq. (119) are immediate and carry no information. The radiated power is ζ times classical by construction, and the time to radiate away the energy of a particle of Lorentz factor γ,

$$t_{\text{cool}} = \frac{\gamma mc^2}{P} = \frac{3m^3c^6}{2\zeta e^4B^2\gamma} \;\propto\; \zeta^{-1}, \tag{120}$$

is inversely proportional to it. Neither is useful on its own, because neither is observable without knowing B and γ independently, which in an astrophysical source one does not.

The observables are the quantities fixed by a *balance*: a particle is accelerated by a field and loses energy by radiating, and what is seen is the equilibrium where the two rates match. That condition is where the ζ dependence stops being trivial, and it is worth doing in general before doing it in cases. Let the accelerating power be eEc, independent of γ, and let the radiated power be

$$P = \zeta\,C\,\gamma^n, \tag{121}$$

with $n = 2$ for synchrotron radiation in a field B and $n = 4$ for curvature radiation on a field line of radius R_c. Equating the two gives the equilibrium Lorentz factor

$$\gamma_{\text{eq}} = \left(\frac{eEc}{\zeta C}\right)^{1/n} \;\propto\; \zeta^{-1/n}, \tag{122}$$

Table 7. How each quantity scales with the radiation-reaction coefficient.

Quantity	Scaling	p	Shift if ζ doubles
Radiated power P	ζ^{+1}	$+1$	$\times 2$
Cooling time $t_{\rm cool}$	ζ^{-1}	-1	$\times 0.50$
Synchrotron burn-off cap $\varepsilon_{\rm max}$	ζ^{-1}	$\mathbf{-1}$	$\times 0.50$
Curvature photon cutoff	$\zeta^{-3/4}$	$-3/4$	$\times 0.59$
$\gamma_{\rm RRL}$, synchrotron at fixed E, B	$\zeta^{-1/2}$	$-1/2$	$\times 0.71$
Curvature-limited γ	$\zeta^{-1/4}$	$-1/4$	$\times 0.84$
$\chi = 0.1$, $\chi = 1$, $B = B_S$ in Fig. 7	ζ^{0}	0	unmoved
$\zeta\tau_0\gamma\omega_B = 1$ in Fig. 7	ζ^{-1}	-1	$\times 0.50$

Note—p is the exponent in $X \propto \zeta^p$. The last two rows are boundaries in the (B, γ) plane rather than observables. The quantum boundaries do not contain ζ at all, so no choice of coefficient can move them.

and if the photon energy characteristic of the emission scales as $\varepsilon \propto \gamma^m$, then

$$\varepsilon \;\propto\; \zeta^{-m/n}. \tag{123}$$

Equation (123) is the result this section exists to establish, and it is worth reading before any numbers are put in. The sensitivity to ζ is not a property of the coefficient. It is the ratio m/n, fixed entirely by how steeply the radiated power and the emitted photon energy each depend on γ. A process whose losses rise steeply with energy is *insensitive* to ζ, because a change in the coefficient is absorbed by a small change in $\gamma_{\rm eq}$; the exponent $1/n$ in Eq. (122) is a dilution factor. Curvature radiation, with $n = 4$, dilutes by a factor of four; synchrotron radiation, with $n = 2$, by two. This single fact decides everything in Sec. 7.3, and it explains why the answer there is split rather than uniform.

7.3. *The synchrotron burn-off cap is linear in ζ; the curvature limits are not*

Putting the cases into Eqs. (122) and (123) gives Table 7, which is the answer to the question Sec. 6.6.3 raised. The exponents are worked out here; the underlying balance conditions are standard and are collected in `REVIEW_2026.md` Secs. 2.6 and 2.8.

One entry deserves to be written out, because it is the only one that is both linear in ζ and free of the quantities an astrophysical source does not hand over. Under ideal magnetohydrodynamics the accelerating field cannot exceed the magnetic field, $E = \eta B$ with $\eta \le 1$. Balancing eEc against Eq. (119) gives $\gamma^2 = 3\eta m^2c^4/2\zeta e^3 B$, and inserting that into the characteristic synchrotron photon energy $\varepsilon = \frac{3}{2}\hbar\gamma^2 eB/mc$ gives

$$\boxed{\varepsilon_{\rm max} = \frac{9}{4}\,\frac{mc^2}{\alpha}\,\frac{\eta}{\zeta} = 158\ \mathrm{MeV} \times \frac{\eta}{\zeta}} \tag{124}$$

Three things about Eq. (124) matter here. The magnetic field has cancelled, so this is a limit that can be applied to a source whose field is unknown, which is most of them. The number is the

Table 8. The same quantities sorted by whether they can be turned into a bound.

Quantity	p	Also needed	Verdict
Damping time, storage ring	-1	nothing; B, γ known	tightest available
Burn-off cap $\varepsilon_{\max}$	-1	η, beaming	weak, but universal
Cooling time, astrophysical	-1	B, γ	degenerate; unusable
$\gamma_{\rm RRL}$	$-1/2$	E, B	unusable
Curvature cutoff	$-3/4$	R_c, E	unusable
Curvature-limited γ	$-1/4$	R_c, E	hopeless

NOTE—Three rows share the exponent $p = -1$ and receive three different verdicts, which is the point of the table: the exponent decides how large the signal is, and the third column decides whether anything else is larger.

familiar synchrotron burn-off limit of high-energy astrophysics, near 160 MeV, recovered here with its dependence on the coefficient made explicit. And the fine-structure constant has appeared again: the cap is $\frac{9}{4}$ of $mc^2/\alpha = 70$ MeV, the same α that Sec. 6.1 identified as the reason the classical theory is never called upon to be consistent and Sec. 6.3.3 as the reason the candidate equations cannot be told apart.

Read in reverse, Eq. (124) is a constraint. Were ζ substantially below unity the cap would lie well above 160 MeV and synchrotron cutoffs should routinely be found in the GeV range, which they are not; were it substantially above, the cap would fall below cutoffs that are observed. The clustering of synchrotron spectral cutoffs near the classical value therefore brackets ζ from both sides. The bracket is an order-of-magnitude one, and Sec. 7.4 says why it is not better than that, but it appears to be the only astrophysical statement about the coefficient available, and it has not been made.

7.4. *The tightest constraint is laboratory and dull; the interesting one is astrophysical and weak*

A large exponent is necessary for a constraint but it is not sufficient, and sorting the rows of Table 7 by p alone gives the wrong answer. A second criterion is needed: what else must be known before the observable can be used at all. Table 8 applies both.

Three quantities share the exponent -1 and receive three different verdicts. The exponent sets the size of the signal; what decides whether the signal is usable is how much else has to be known to see it. Figure 9 draws that statement: the three loci respond equally strongly to ζ and differ only in shape and in whether a prior truncates them.

The tightest bound is already in hand, and it is in an accelerator. Storage rings are easiest for a prosaic reason: the quantities that are unknown in an astrophysical source are machine settings in a ring. The beam energy fixes γ, the dipole lattice fixes ρ or B, the revolution period T_0 is measured, and the damping partition numbers J_i are determined by the lattice. The quantum correction is also absent for practical purposes: Sec. 6.5.1 put a typical ring at $\chi \sim 10^{-6}$, where $g(\chi)$ departs from unity by only one part in 10^5 and cannot be mistaken for a change in ζ.

What is measured is not an abstract cooling time in the astrophysical sense. The direct observables are damping decrements: the exponential decay of betatron and synchrotron oscillation amplitudes

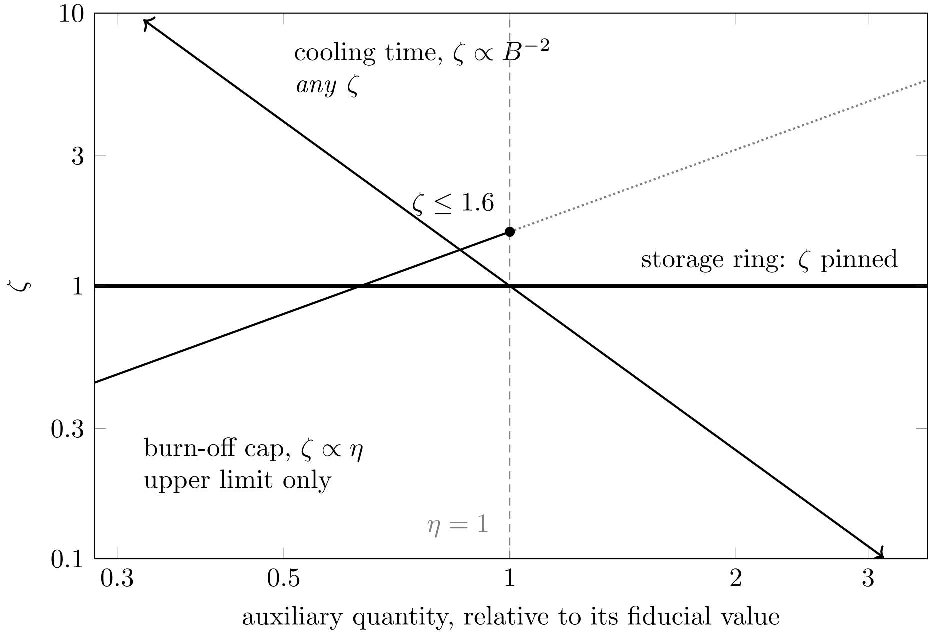


Figure 9. Why three observables with the same exponent give three different verdicts. Each locus is the set of parameter pairs consistent with *one* measurement; the horizontal axis is whatever must be known first and is not, namely nothing for the ring, η for the cap, B for a cooling time. All three scale as ζ^{-1} and so respond equally; only the shape differs. The ring's locus is horizontal and fixes ζ; the cap's has slope one and is cut at $\eta = 1$, which removes one side only, leaving an upper limit (the marked $\zeta \leq 1.6$ is for a 100 MeV cutoff); the cooling time's has slope -2 with no prior to cut it, so every ζ survives. *The exponent sets how strongly an observable responds; the geometry of its locus decides whether the response can be attributed to* ζ*.*

after injection or after a deliberate kick. With the coefficient made explicit, Eq. (109) becomes

$$\tau_i(\zeta) = \frac{2E_0T_0}{J_i\,\zeta U_{0,\mathrm{cl}}}, \qquad \frac{1}{\tau_i} = \zeta\,\frac{J_iU_{0,\mathrm{cl}}}{2E_0T_0}. \tag{125}$$

Here $U_{0,\mathrm{cl}}$ is the classical energy loss per turn computed with $\zeta = 1$. Thus the measured slope of $\ln A_i$ against time is linear in ζ. The same coefficient also appears in the RF power needed to replace the turn-by-turn energy loss, but the damping decrement is the cleanest way to say what the machine measures.

There are already data of this kind, although they were not advertised as measurements of ζ. The SLC damping-ring measurements, for example, determined the synchrotron-radiation energy loss per turn from the dependence of the synchronous phase on RF voltage. At $E = 1.21$ GeV they found $U_0 = 84.0$ keV, to be compared with the calculated 84.4 keV, and inferred a vertical damping time of 3.4 ms against a calculated 3.45 ms (L. Rivkin et al. 1985). Other storage rings have measured damping rates as machine parameters, including their dependence on bunch current, chromaticity, and collective effects (K. T. Hsu et al. 1995). These are measurements of radiation damping, but their purpose was accelerator physics: commissioning, impedance, emittance, and stability.

That is why the answer to "why has no one measured this?" is a little misleading. Damping has been measured many times. What has not usually been done is to isolate a low-current, low-

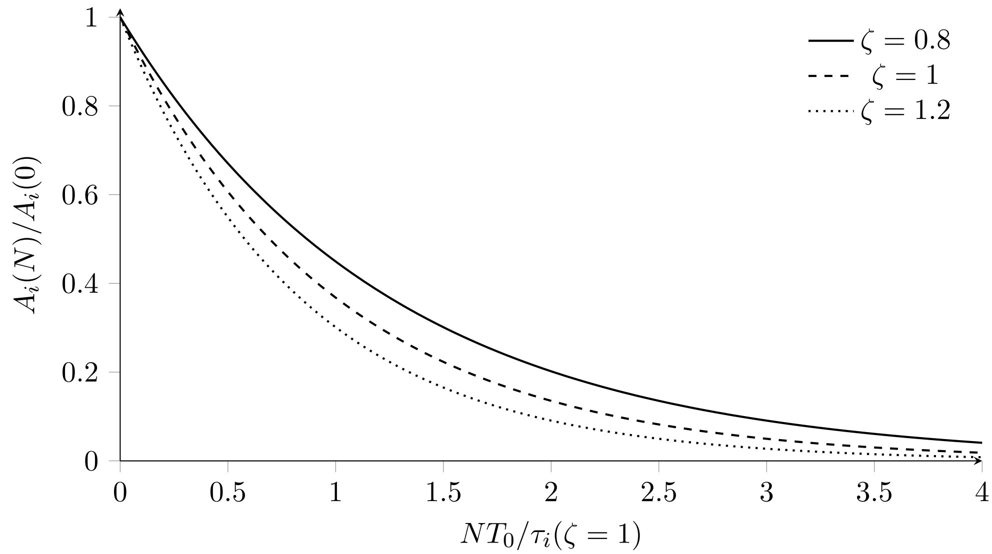


Figure 10. What a storage-ring damping measurement reduces to in its simplest form. The machine records turn-by-turn beam motion or an equivalent diagnostic; the analysis extracts an oscillation envelope $A_i(N) = A_i(0)\exp[-NT_0/\tau_i(\zeta)]$. The curves are not raw data. They are the exponential fits, or predictions, whose slope gives the damping decrement. Changing ζ changes that slope and does not require an independent estimate of an astrophysical field or particle energy.

collective-effect data set and publish the same comparison as a bound on

$$\zeta = \frac{\text{measured damping decrement}}{\text{classical damping decrement predicted from the machine model}}.$$

For accelerator operation this ratio is bookkeeping. For the present question it is the cleanest laboratory null test of the coefficient multiplying radiation reaction.

This is the item Sec. 6.6.3 called bookkeeping, and the exponent analysis shows it is not merely the easiest constraint but the strongest. An agreement between measured and predicted damping decrements at the per cent level is a per cent bound on ζ, and seventy years of such agreements exist. The result would be $\zeta = 1$ to a precision no astrophysical argument can approach, and its interest lies entirely in the fact that the number has never been written down.

The interesting bound is astrophysical and it is an order of magnitude. Equation (124) is degenerate with η and with Doppler boosting, and both are large effects: the Crab flares reach about 400 MeV, which is $\eta/\zeta \approx 2.5$, and are attributed to reconnection regions where $E > B$ or to beaming, either of which accounts for the excess without touching the coefficient. A single source therefore says nothing. What does say something is the asymmetry between the two kinds of quantity. **ζ is a constant of electrodynamics and is the same in every source; η and the Doppler factor are properties of a particular geometry and differ from one source to the next. A population of independent sources whose synchrotron cutoffs cluster near the classical value therefore bounds ζ in a way that no per-source systematic can imitate, because a systematic would have to take the same value everywhere to do so.** That is the form the astrophysical statement should take, and it is a statement about a sample rather than about an object.

The rest is dead. Curvature radiation dilutes by the factor of four of Eq. (122) and additionally requires the field-line radius and the accelerating field, neither of which is measured; $\gamma_{\rm RRL}$ requires E and B separately. These were the routes the original magnetar motivation suggested, and Table 8 closes them. The regions of Fig. 7 that remain usable are the low-field magnetospheric ones, $B \ll B_S$ and $\chi \ll 1$; the magnetar surface is an exclusion zone for the reasons of Sec. 7.7 and not a place where any of this applies.

The shape of the answer is therefore not the one Sec. 7 warned might come out — it is not that everything dilutes as $\zeta^{1/4}$ — but it is not a clean success either. The constraint that can be made precise is one nobody will find surprising, and the constraint that would be interesting is one that can only be made loosely. Both are worth having, and for different reasons: the first because a mature subject should be able to quote the precision of its own central coefficient, and the second because it is the only place where the coefficient is exposed to data taken outside a laboratory.

7.5. *In the transition region ζ and $g(\chi)$ multiply, and no one has separated them*

[The second and more interesting half of Sec. 6.6.3, developed rather than restated. The observable is $P_{\rm rad} = \zeta\, g(\chi) P_{\rm cl}$. Every strong-field result of Sec. 6.5 constrains g at fixed $\zeta = 1$. Two questions: whether the χ dependence of g breaks the degeneracy in principle — it should, since ζ is a constant and g is a known function, so a scan in χ separates them while a single operating point cannot — and whether the existing data span enough χ to do it. This is the argument for why a χ-scanning machine (Sec. 6.5.8) matters for the coefficient and not only for the rates. Requires no new data to start.]

7.6. *The same diagram audits where the classical description is used outside its domain*

[Figure 7 is not decoration but a test that takes one number. Compute $\chi = \gamma B_\perp / B_S$ before applying any classical radiation-reaction treatment; if $\chi \gtrsim 0.1$ the classical radiated power is wrong by tens of per cent (Sec. 6.4) and by 82% at $\chi = 1$. The worked example is the author's own: the 2010 and 2011 drafts in this repository assume a condition that places them at $\chi \approx 200$, some 2.3 decades inside the quantum region, as `REVIEW_2026.md` Sec. 2.6(d) computes. Using one's own work as the example is the honest way to raise the general question of how often this happens elsewhere, and it should be raised as a question, not asserted as a finding, until the literature has been checked.]

7.7. *What is not claimed*

[Short, and necessary because the preceding subsections invite overreading. Not a revival of the λ–κ family: ζ is a null-test parameter and $\zeta = 1$ remains the prediction (Sec. 6.3.4). Not a reopening of the classical equation of motion, which Secs. 6.1 to 6.3 treat as settled. Not a claim that magnetar surface fields revive classical theory — they are an exclusion zone and the original 10^{15} G motivation of D. Zhang (2006) has to be restated as a strong-field QED problem. And the principal risk stated plainly: ζ may decohere against collision geometry and beam-parameter systematics badly enough that no useful interval survives.]

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